

Machine Translation

Prof. Luca Cagliero
Dipartimento di Automatica e Informatica
Politecnico di Torino



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di Torino**

Lecture goal

- Preliminaries
- Rule-based approaches
- Statistical Machine Translation
- Neural Machine Translation

Machine translation

- Automatic translation of a sentence from a source language to a target language



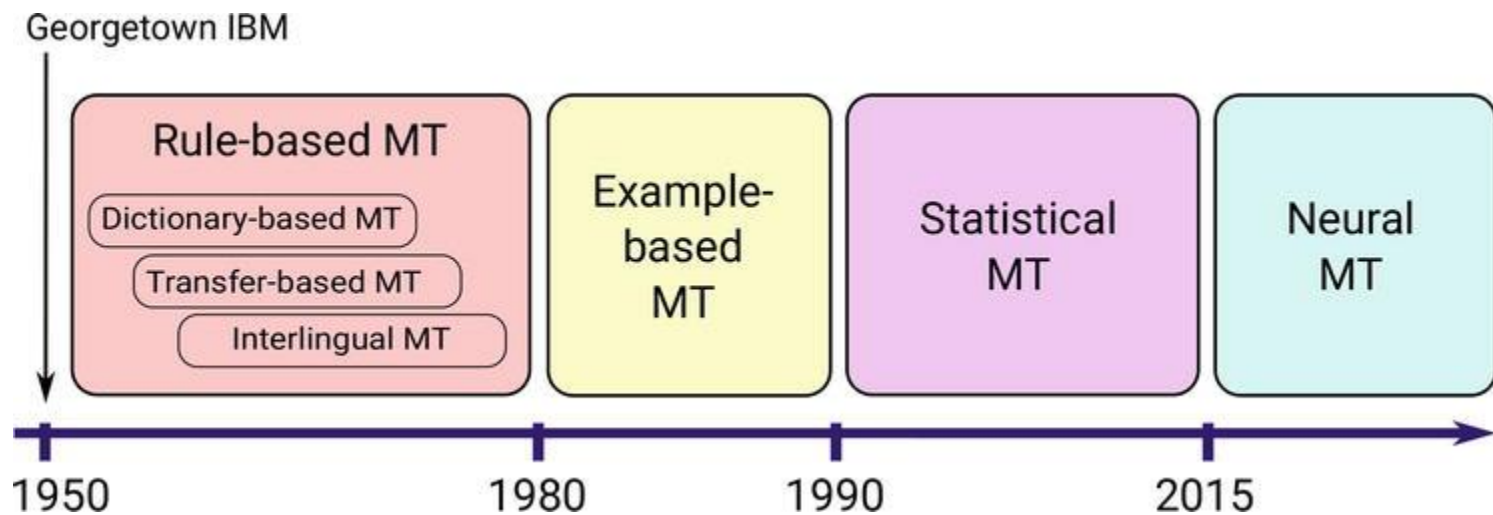
Machine translation

- Approaches
 - Rule- or dictionary-based
 - Statistical
 - Neural



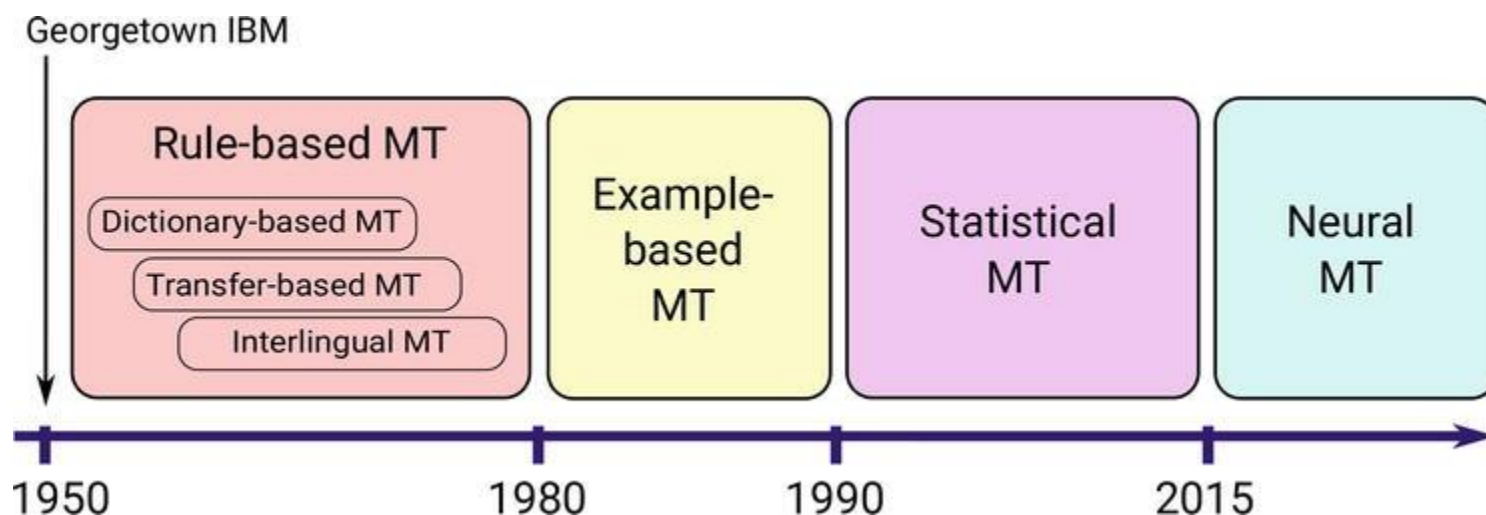
Rule-based Machine Translation

- Machine translation systems based on linguistic information about source and target languages basically retrieved from (bilingual) dictionaries and grammars
- Also known as **Knowledge-Based Machine Translation** or **Classical MT Approach**



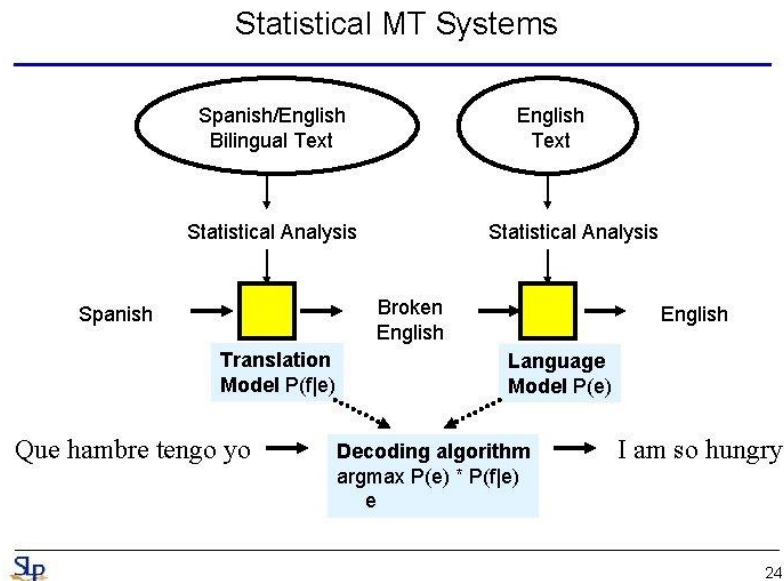
Example-based Machine Translation

- Bilingual corpus with parallel texts
 - set of sentences in the source language (from which one is translating)
 - the corresponding translations of each sentence in the target language with point-to-point mapping
- Translate by analogy



Statistical Machine Translation

- Based on statistical models whose parameters are derived from the analysis of bilingual text corpora.
- Key ideas
 - every sentence in one language is a possible translation of any sentence in the other
 - the most appropriate is the translation that is assigned the highest probability by the system



Statistical Machine Translation

- Learn a probabilistic model from data
- Given a sentence x in the source language find the best sentence y in the target language

$$\arg_y \max P(y \mid x)$$

- Apply Bayes theorem

$$\arg_y \max P(x \mid y)P(y)$$

$P(y)$: language model

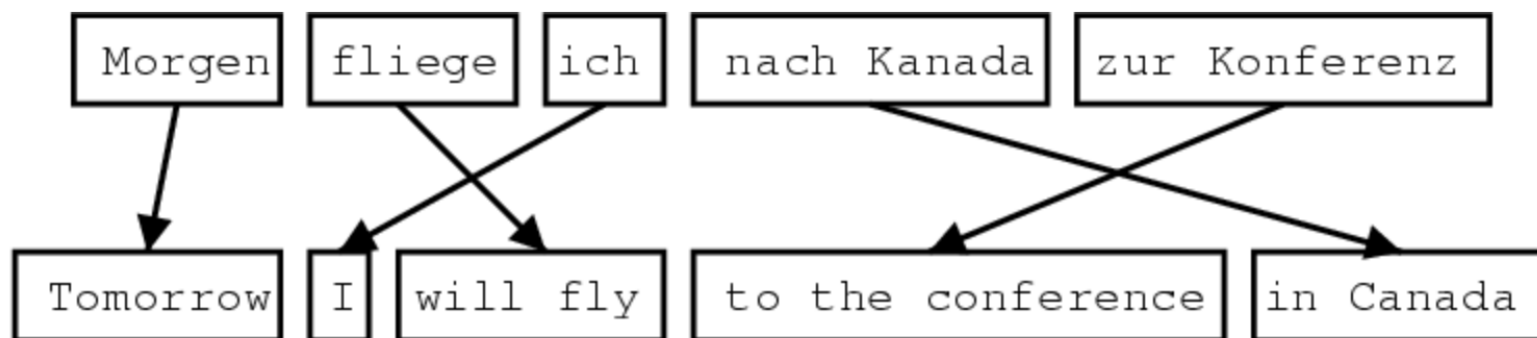
$P(x \mid y)$: translation model

Statistical Machine Translation

- The **translation model** describes how words and phrases should be translated
 - Requires a bilingual collection of human-translated sentences
- The **language model** describes how to write in the target language
 - Requires a monolingual collection

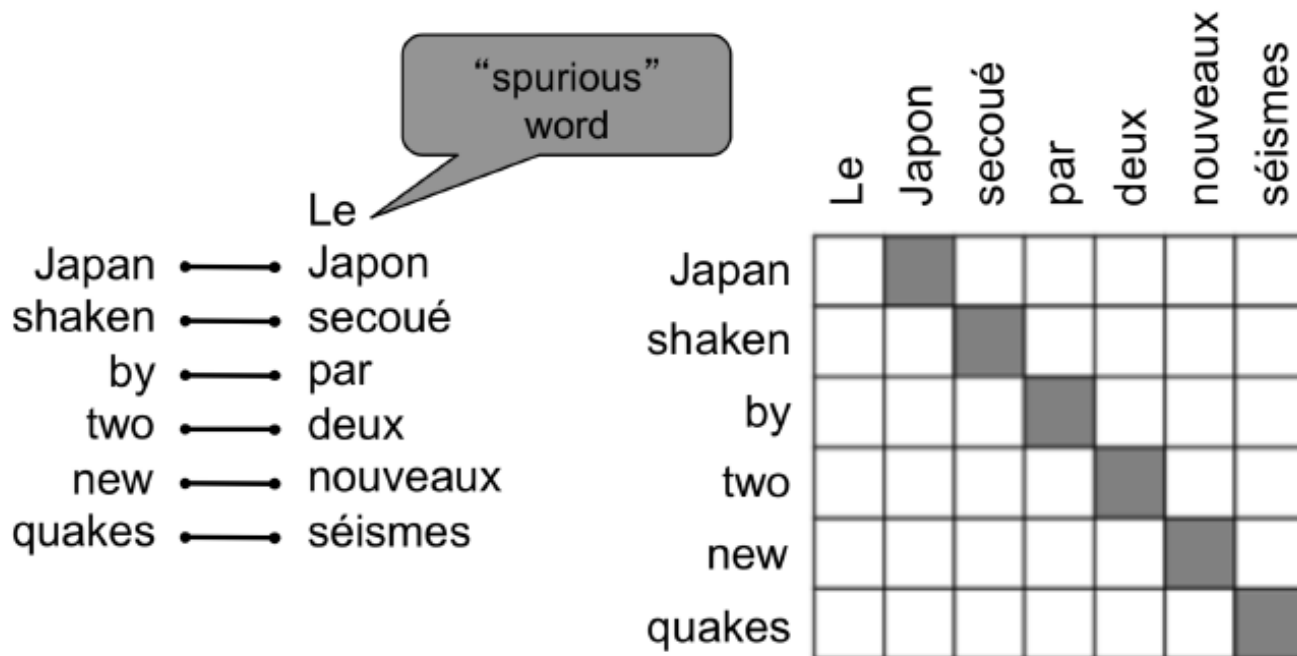
Translation model learning

- $P(x \mid y) = P(x, a \mid y)$
- a: alignment term
 - word-level correspondence between source sentence x and target sentence y



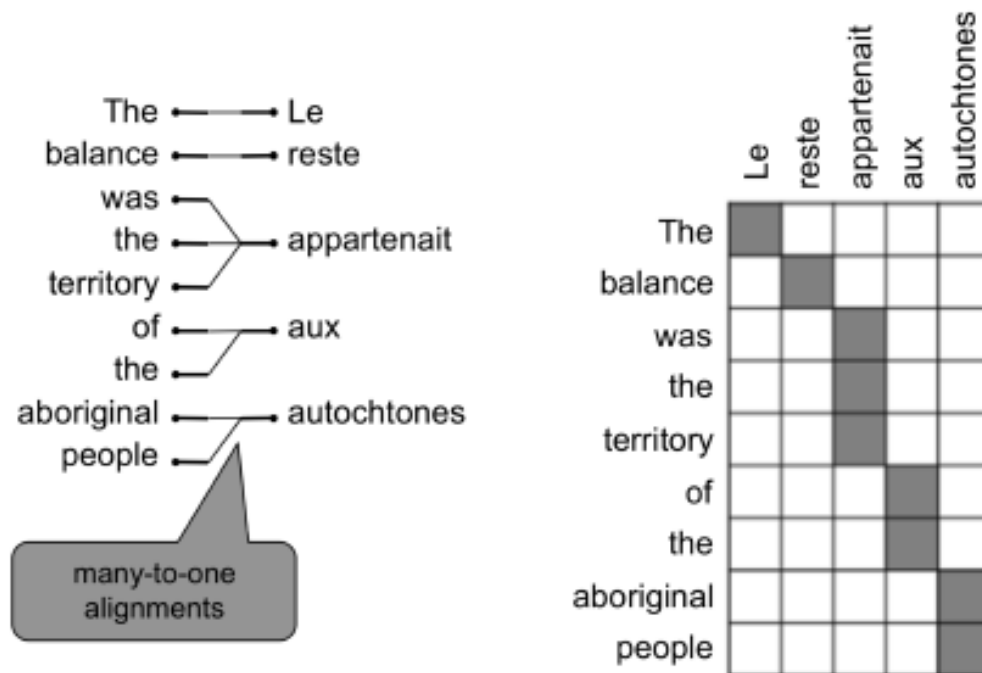
Alignment

- Bilingual lexicon
 - correspondence between particular words
 - often incomplete



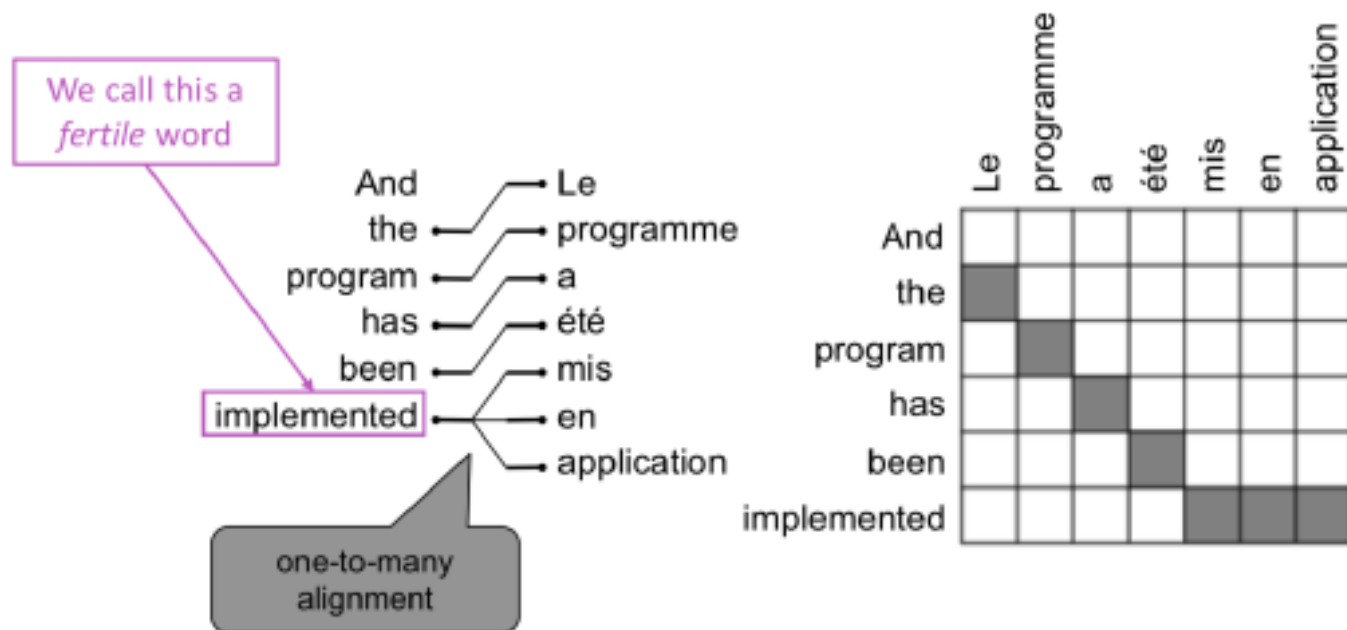
Alignment

- **Many-to-one** correspondences are allowed



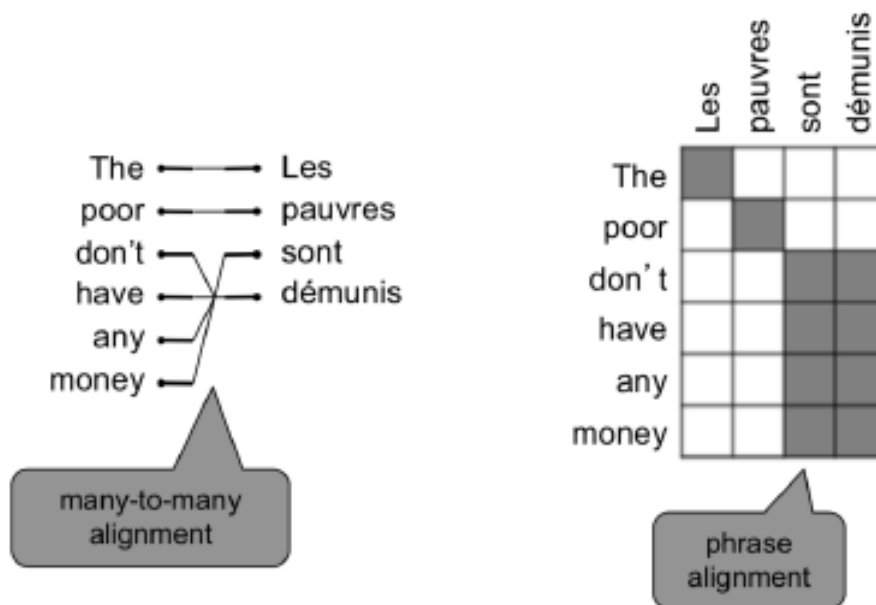
Alignment

- **One-to-many** correspondences are allowed



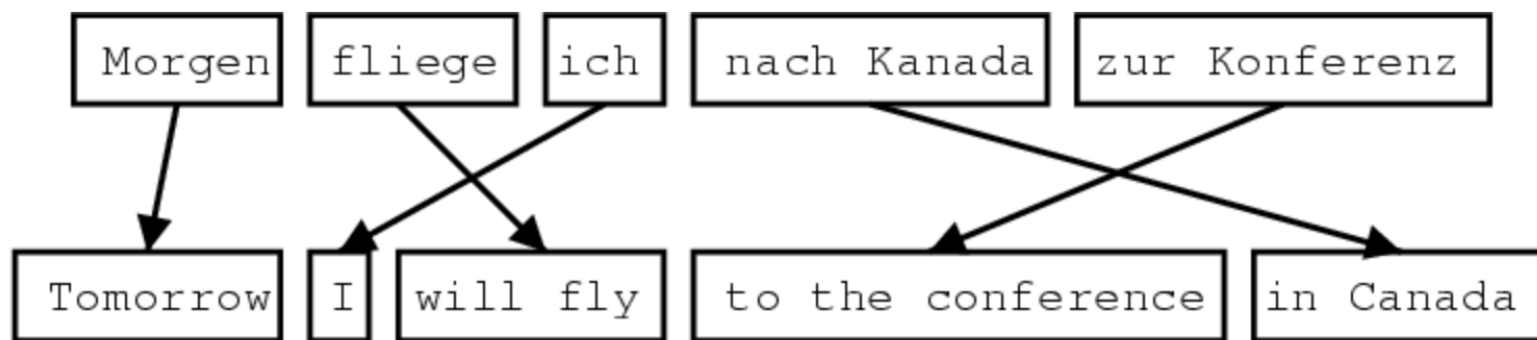
Alignment

- **Many-to-many** correspondences are allowed



Translation model learning

- $P(x, a \mid y)$ is influenced by
 - Probability of word alignment
 - Text position
 - ...
- a is a latent variable
 - Expectation-Maximization



Decoding

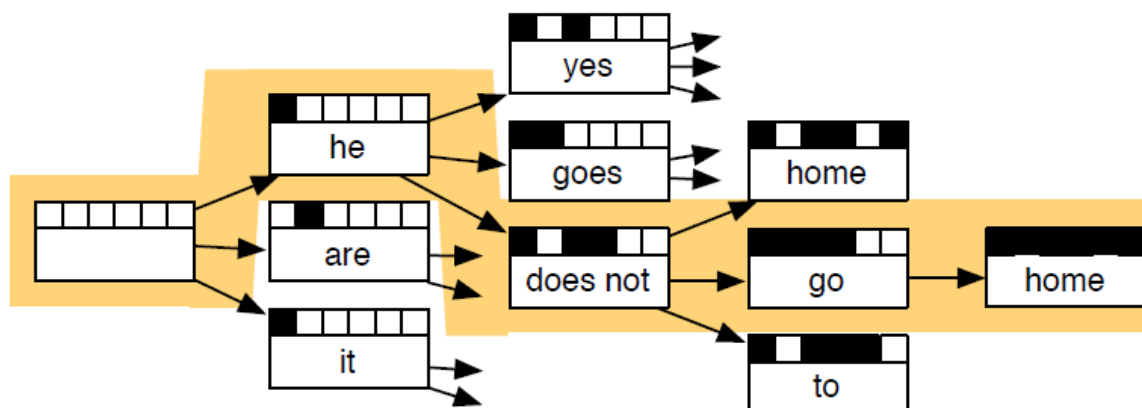
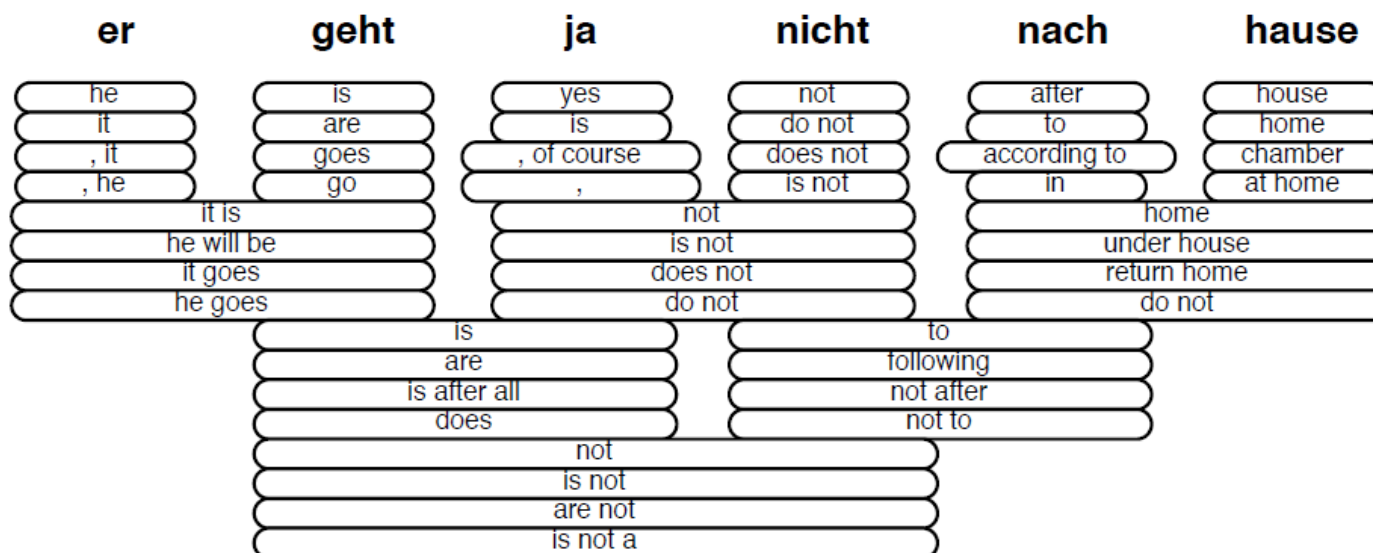
- Maximization process

$$\arg_y \max P(x | y)P(y)$$

- Unfeasible without enforcing independence assumptions

Decoding

- Tree-based candidate exploration and evaluation



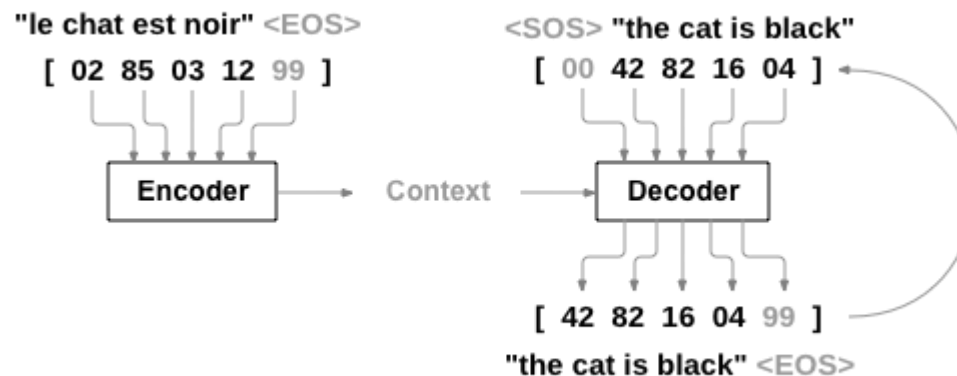
Koehn, 2009 Statistical Machine Translation

Challenges of Statistical Machine Translation

- Lots of feature engineering
 - Hand-crafted features used to capture peculiar language phenomena
- Limited scalability
 - Requires a large knowledge base
 - E.g., tables of equivalent phrases
- Lots of human efforts
 - Model maintenance
- Very limited portability
 - Ad hoc for a source/target language pair

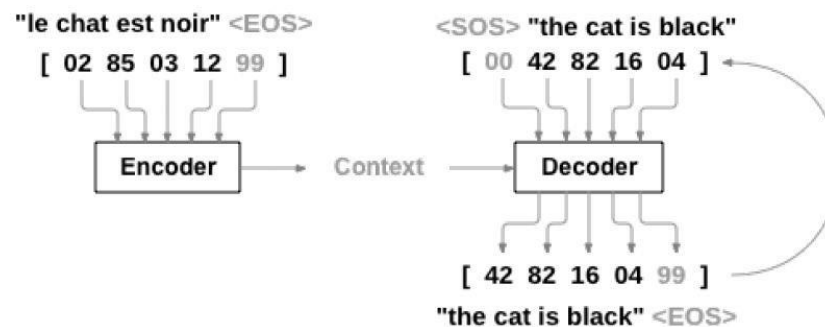
Neural Machine translation

- End-to-end Neural Network training on a large-scale bilingual corpus
 - Leverage Deep Learning architectures
- Sequence-to-sequence models (seq2seq)
 - End-to-End backpropagation

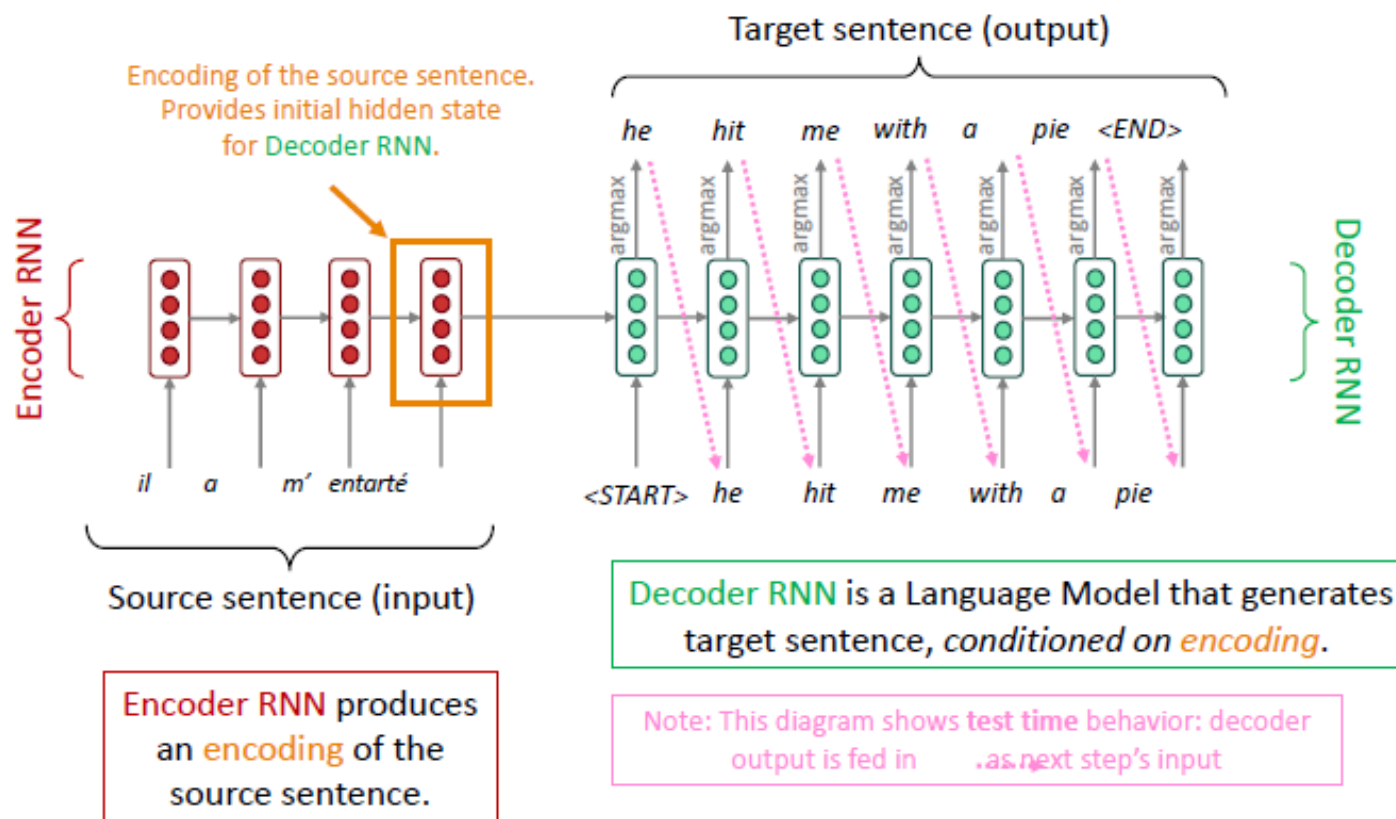


Neural Machine translation: the encoder decoder architecture

- Input and output are both variable-length sequences
- Encoder: it transforms the input into the encoded vector
 - The Encoder RNN produces an encoding of the source sentence
 - The vector encapsulates the information for all input elements in order to help the decoder make accurate predictions
- Decoder: maps the encoded state of a fixed shape to a variable-length sequence
 - The Decoder RNN is a Language Model that generates the target sentences conditioned on encoding
 - It codes the state to generate the translated sequence token by token as the output



Seq2Seq



Seq2Seq

- Seq2Seq models are suitable for various NLP tasks
 - **Machine Translation**
 - From source sentence to target sentence
 - **Summarization**
 - from long text to shorter text
 - **Dialogue**
 - from the previous utterance to the next one
 - **Parsing**
 - from the input raw text to the parsed output
 - **Code generation**
 - from natural language to a programming language

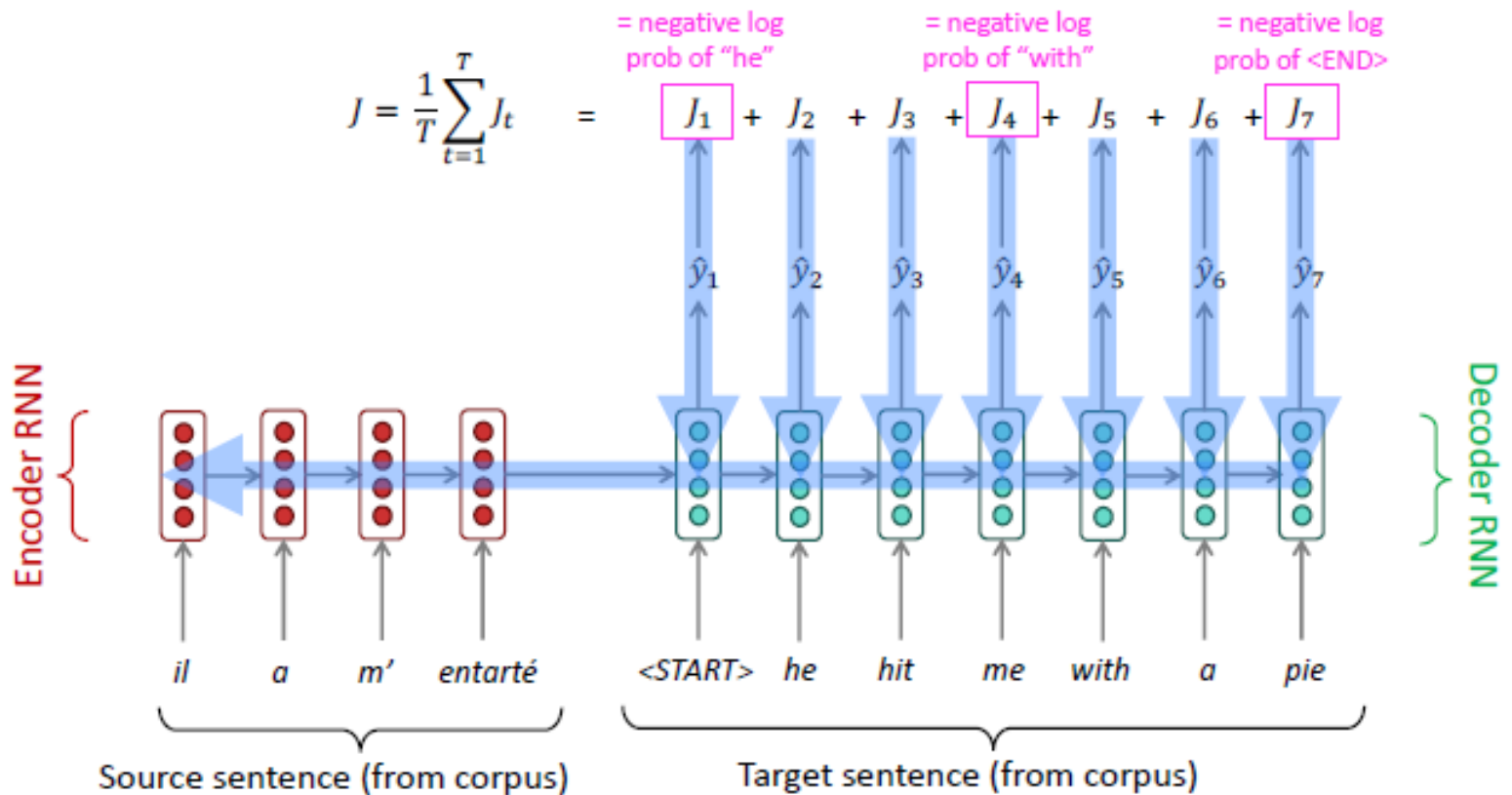
Seq2Seq

- Seq2Seq is a conditional language model
 - **Language model**
 - The decoder predicts the next word of the target sentence
 - **Conditional**
 - Predictions are conditioned on the source sentence

$$P(y|x) = P(y_1|x) P(y_2|y_1, x) P(y_3|y_1, y_2, x) \dots \underbrace{P(y_T|y_1, \dots, y_{T-1}, x)}$$

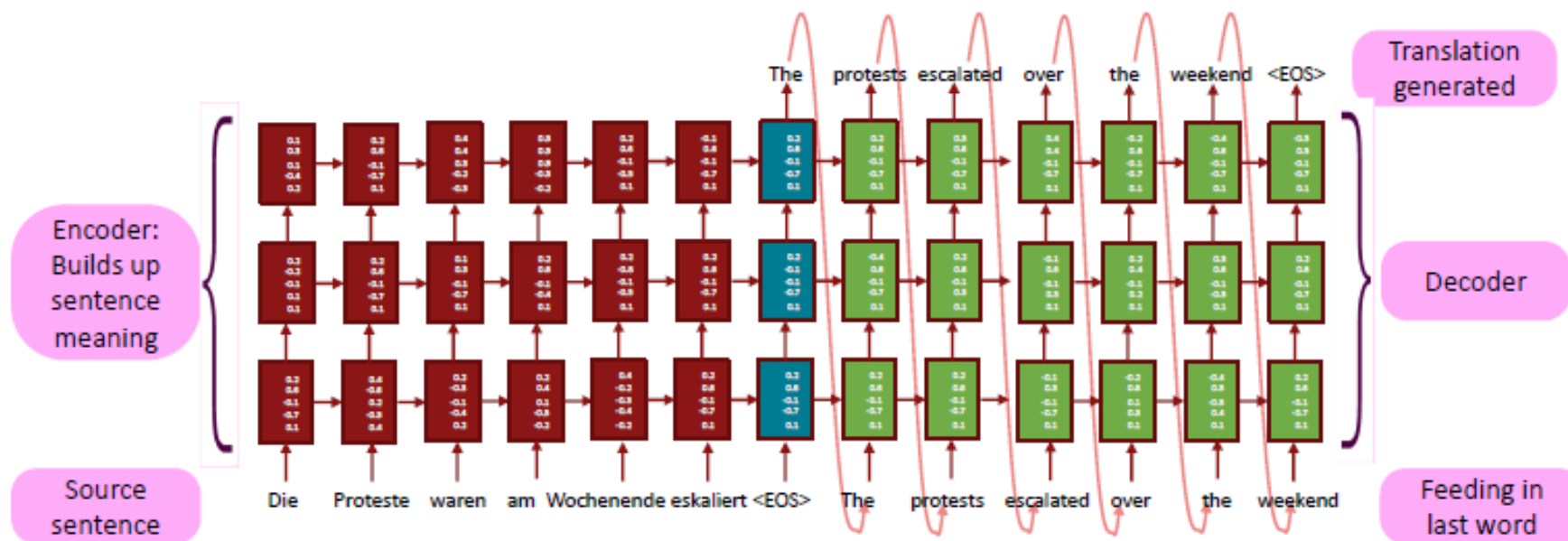
Probability of next target word, given target words so far and source sentence x

Seq2Seq: end-to-end optimization



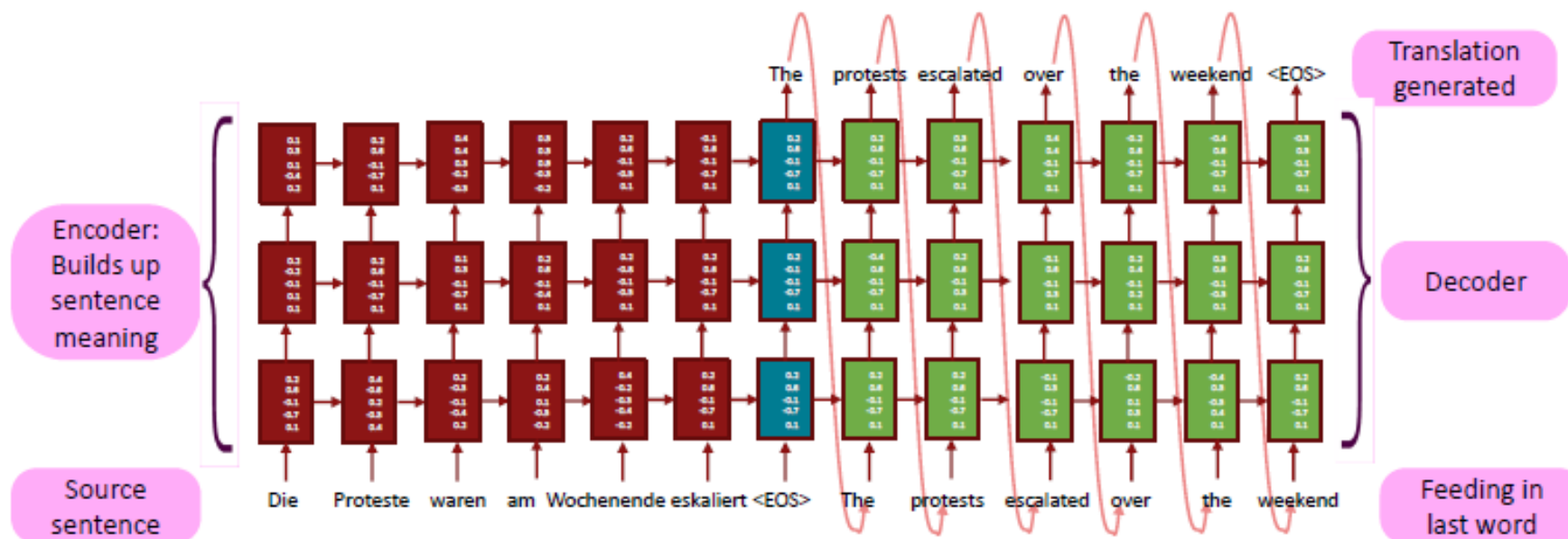
Multi-layer Encoder-Decoder

- Stacked RNNs
 - “Deep” not only in the temporal dimension
 - Apply multiple RNNs
 - The hidden states from RNN layer i are the input of RNN layer $i+1$



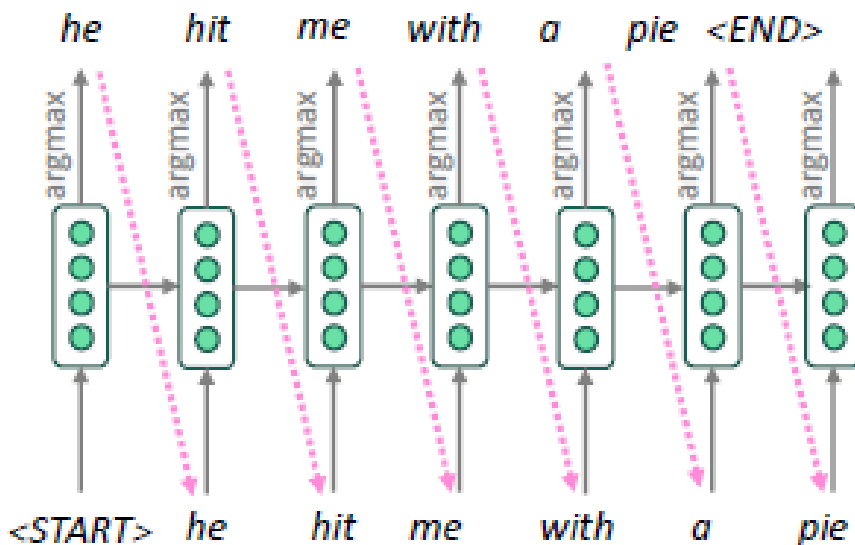
Multi-layer Encoder-Decoder

- Stacked RNNs
 - The lower RNNs compute lower-level features
 - The higher RNNs compute higher-level ones



Greedy encoding

- Decode the target sentence by taking arg max on each step of the decoder
 - Take the most likely word on each step
 - No way to undo decisions



Exhaustive encoding

- Find the length- T best translation by maximizing

$$\begin{aligned} P(y|x) &= P(y_1|x) P(y_2|y_1, x) P(y_3|y_1, y_2, x) \dots, P(y_T|y_1, \dots, y_{T-1}, x) \\ &= \prod_{t=1}^T P(y_t|y_1, \dots, y_{t-1}, x) \end{aligned}$$

- Compute all the possible sequences
 - Track all possible partial translations
 - Computationally intractable!
 - $O(V^T)$, where V is the vocabulary size

Beam search encoding

- Keep track of the k most likely translations
 - K is the beam size (typically, between 5 and 10)
- Define the hypotheses $y_1, \dots, y_t$ and their scores
- Log probability

$$\text{score}(y_1, \dots, y_t) = \log P_{\text{LM}}(y_1, \dots, y_t | x) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

- Search for the top ranked hypotheses by considering only the top- k translations at each step

Beam search encoding

- No guarantee to find the optimal solution
- Efficiency is significantly higher than exhaustive encoding
- Key steps
 - Calculate the probability distribution of the next word
 - Take k words and compute the scores
 - For each of the k hypothesis find top- k next words and calculate the scores
 - Among the K^2 available hypotheses keep the k hypothesis with highest score
 - Backtrack to obtain the full hypothesis
 - Stop decoding when the model produces an $\langle \text{END} \rangle$ token
 - Alternative stopping criteria
 - Until a predefined timestep T is reached
 - Until at least a predefined number n of hypotheses have been completed

Beam search encoding

- Each hypotheses has a score

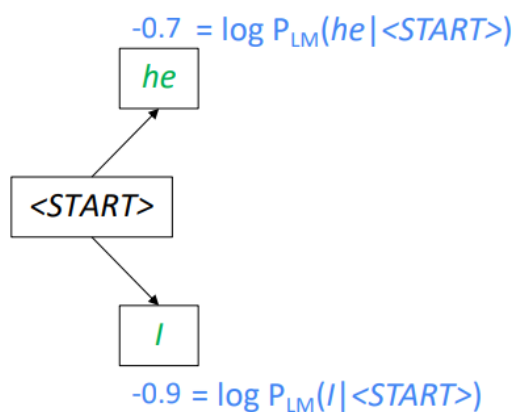
$$\text{score}(y_1, \dots, y_t) = \log P_{\text{LM}}(y_1, \dots, y_t | x) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

- Longer hypotheses have lower scores
 - Normalize by length:

$$\frac{1}{t} \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

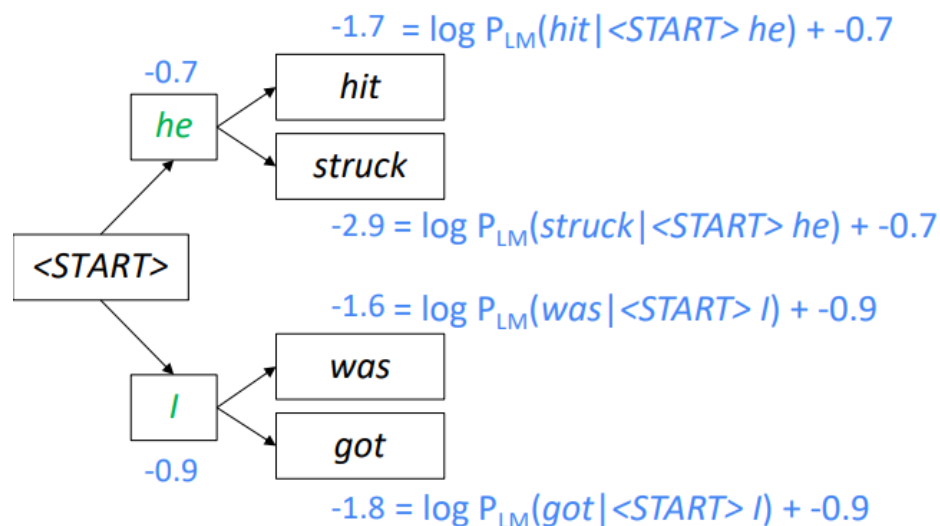
Beam search encoding

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



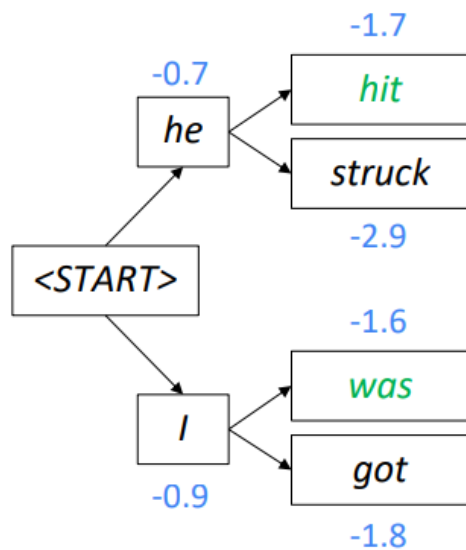
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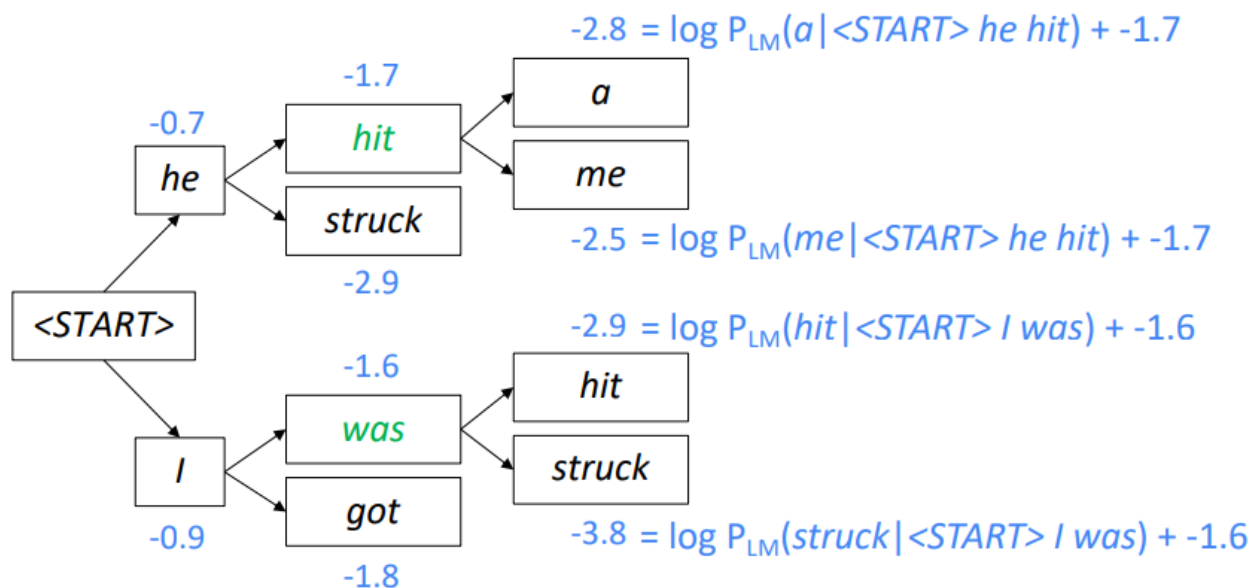
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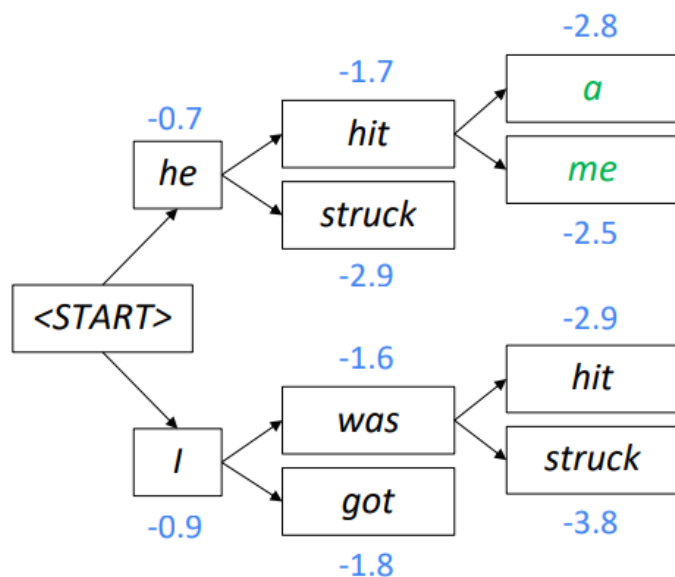
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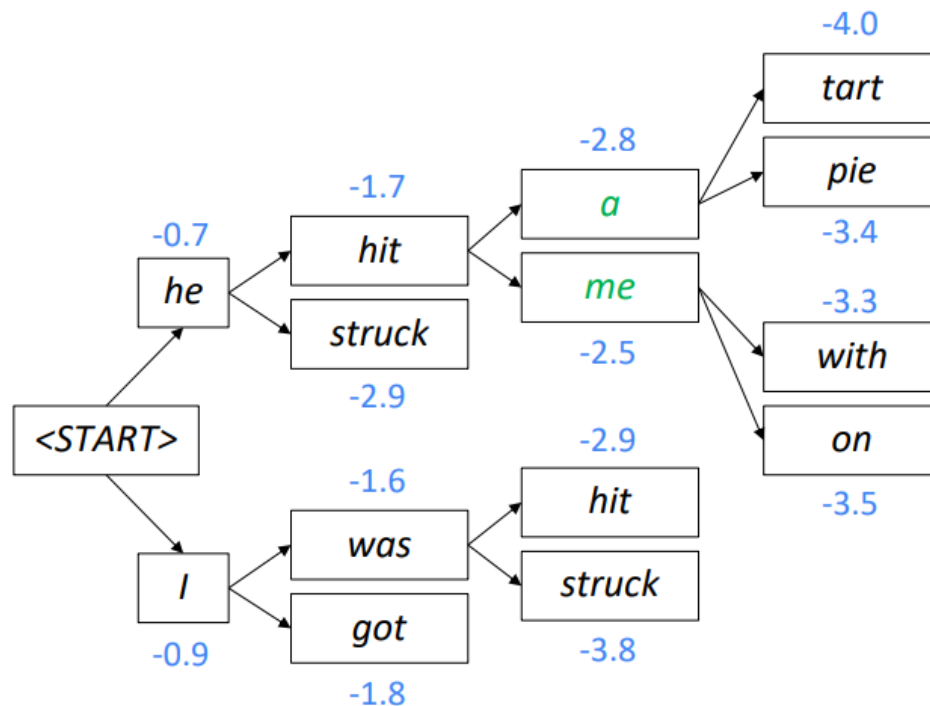
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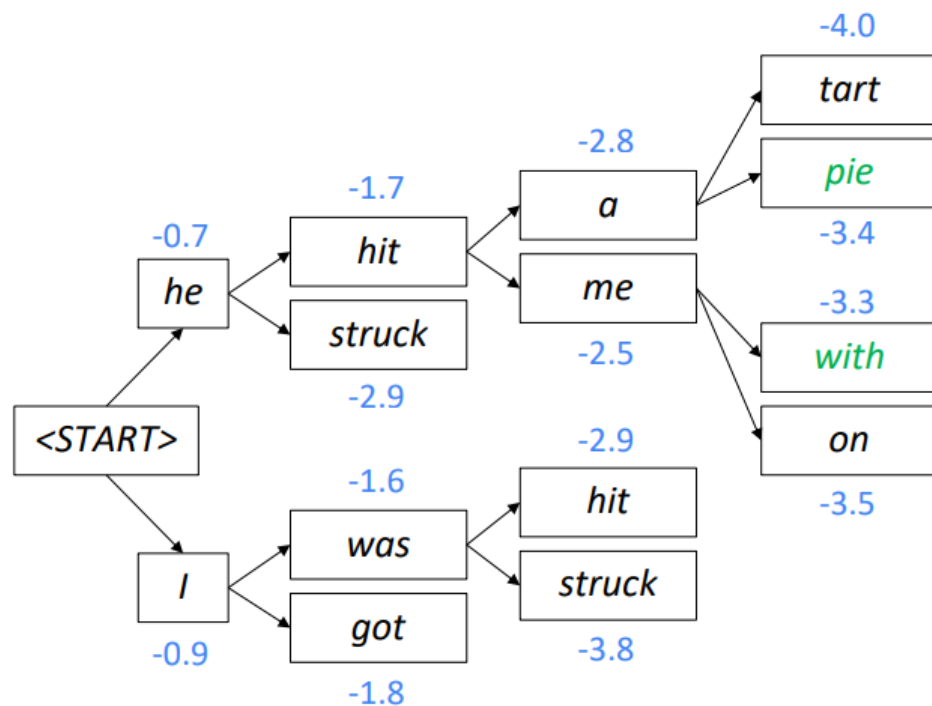
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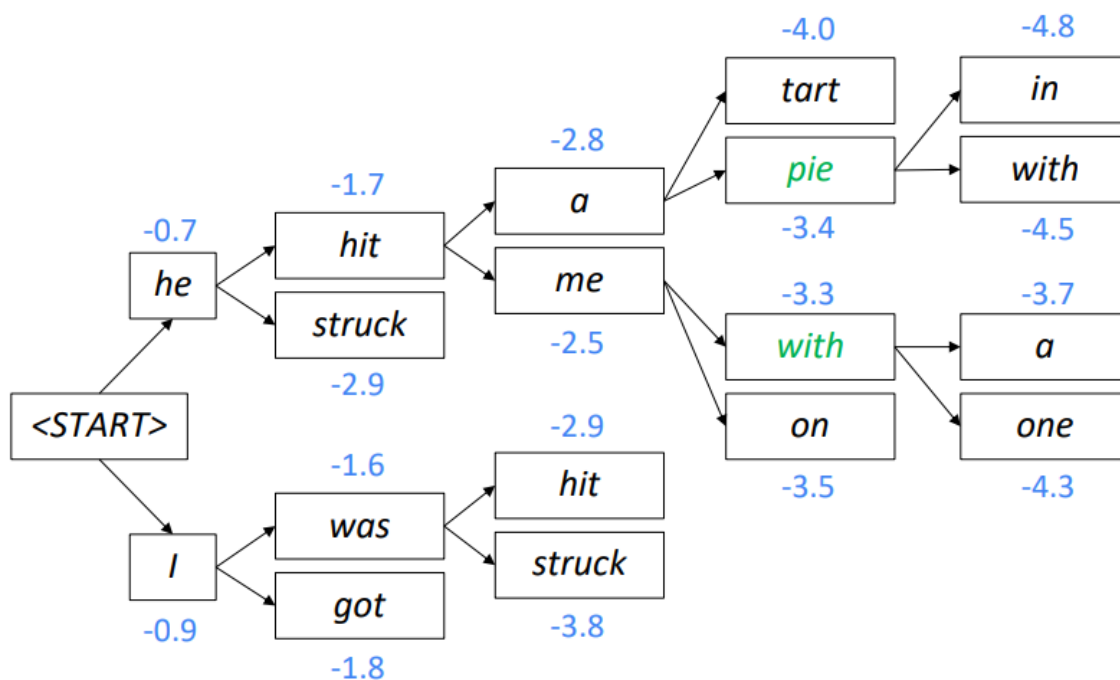
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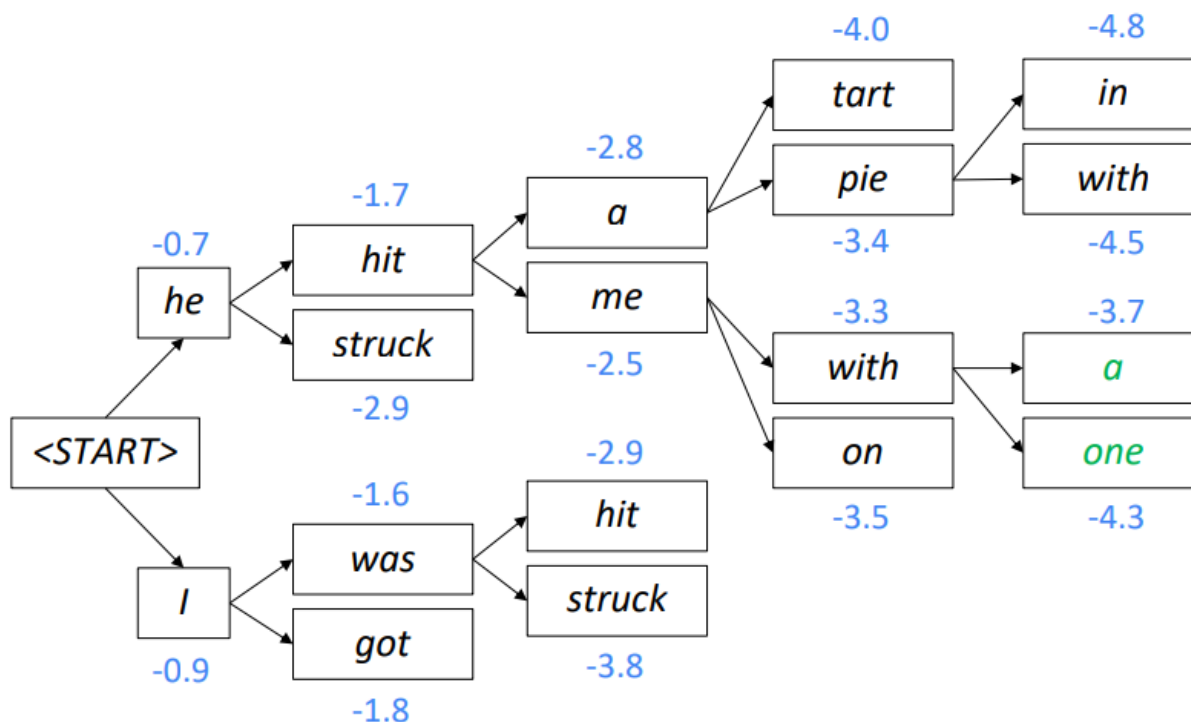
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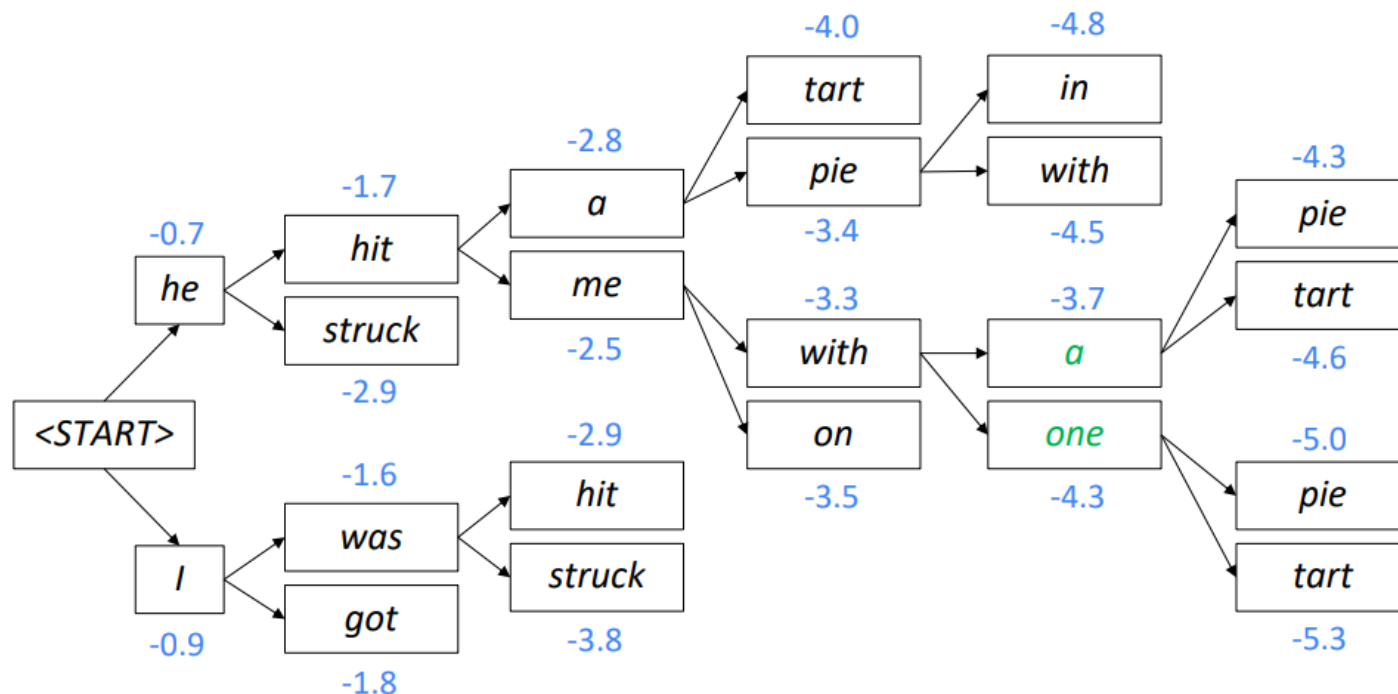
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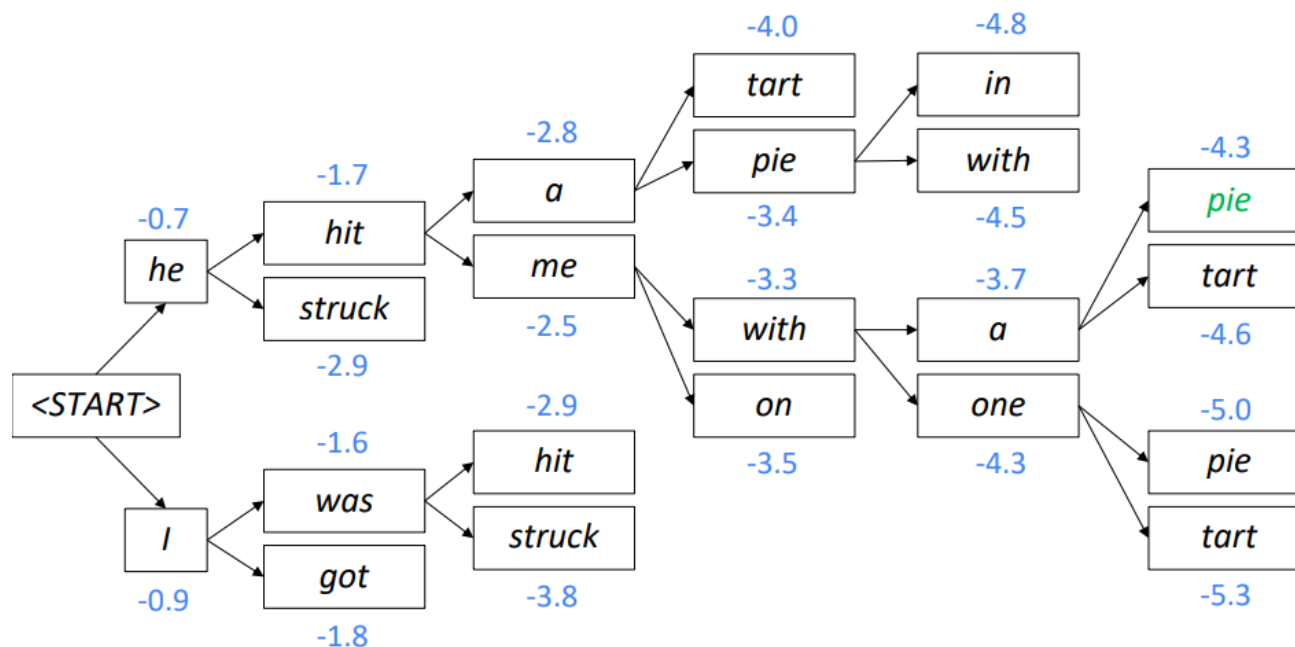
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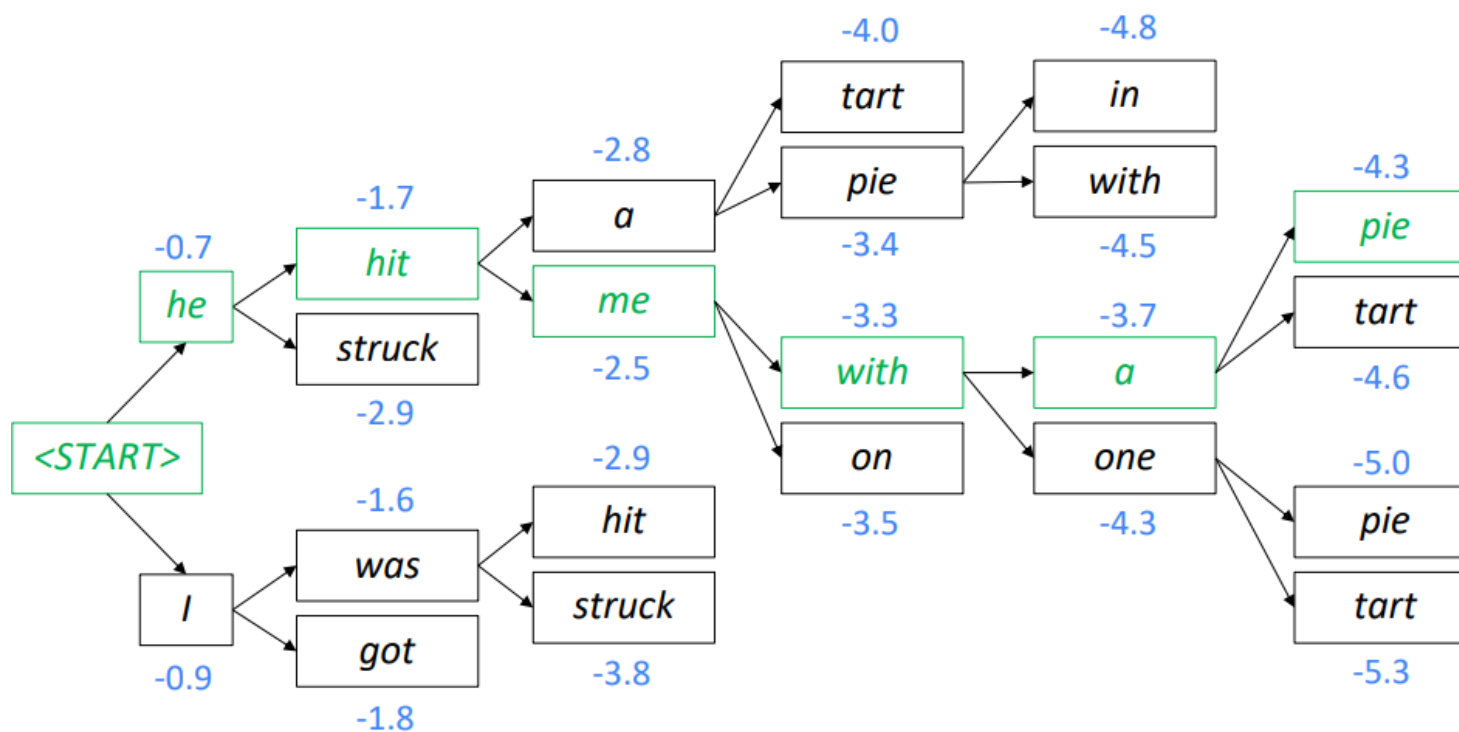
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<START> he hit me with a pie <END>

Bilingual Evaluation Under study (BLEU)

- Metric to compare the machine-generated translation to one or more human-written summaries
 - Based on co-occurrence-based metrics
 - N-gram precision
 - Hypothesis: a single translation per person
 - Low N-gram overlap with the human translation

Bilingual Evaluation Under study (BLEU)

- Compute the geometric average of the modified n-gram precisions p_n using n-grams up to length N and positive weights w_n summing to one
 - Baseline: $N=4$, $w_n = 1/N$

$$p_n = \frac{\sum_{C \in \{Candidates\}} \sum_{n\text{-gram} \in C} Count_{clip}(n\text{-gram})}{\sum_{C' \in \{Candidates\}} \sum_{n\text{-gram}' \in C'} Count(n\text{-gram}')}$$

Bilingual Evaluation Under study (BLEU)

- Let c be the length of the candidate translation and r be the effective reference corpus length
- Compute the brevity penalty BP

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

Bilingual Evaluation Under study (BLEU)

- Then

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

- The ranking is usually computed in the log domain

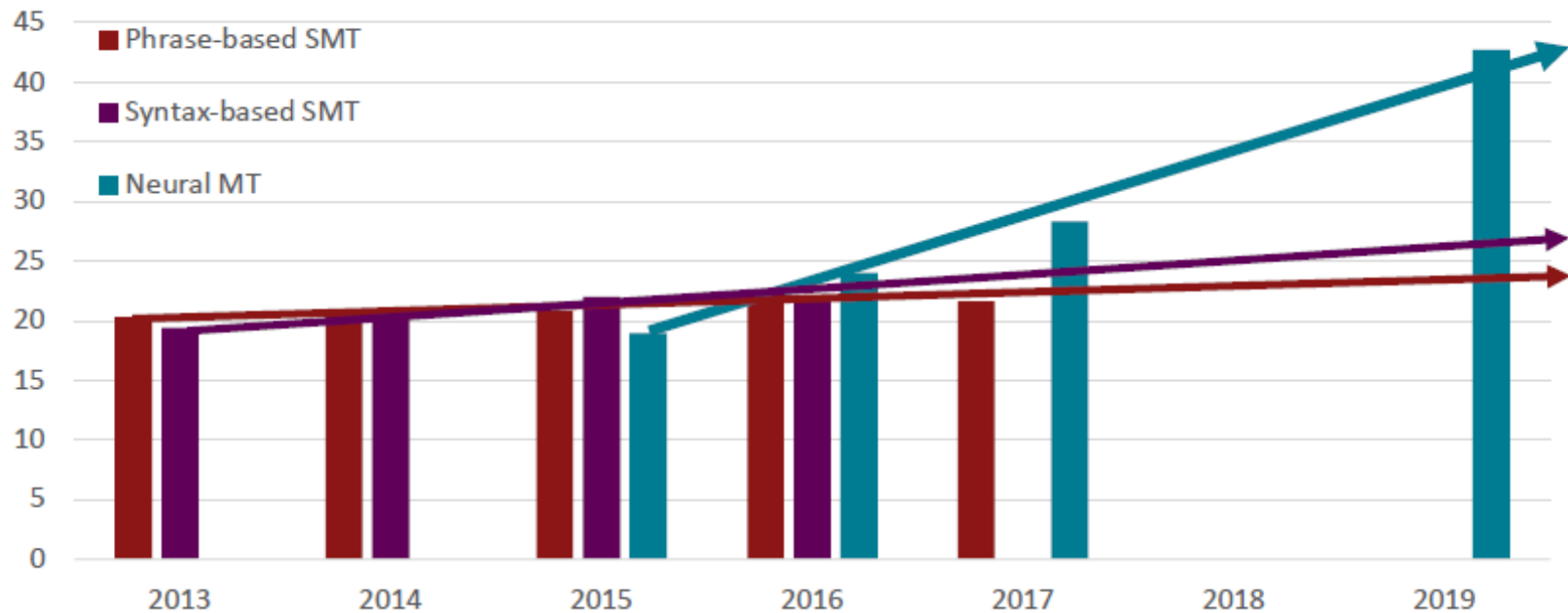
$$\log \text{BLEU} = \min\left(1 - \frac{r}{c}, 0\right) + \sum_{n=1}^N w_n \log p_n.$$

Additional reading on BLEU



- Kishore Papineni, Salim Roukos, Todd Ward, Wei-Jing Zhu. BLEU: a Method for Automatic Evaluation of Machine Translation. ACL 2002: 311-318
- Download and read the paper: <https://aclanthology.org/P02-1040.pdf>

Machine Translation evolution over time



Pros of Neural Machine Translation

- Text fluency
- Better contextualization
- No sub-components to be optimized
 - End-to-end training
- Limited human engineering effort
- Same method for all language pairs

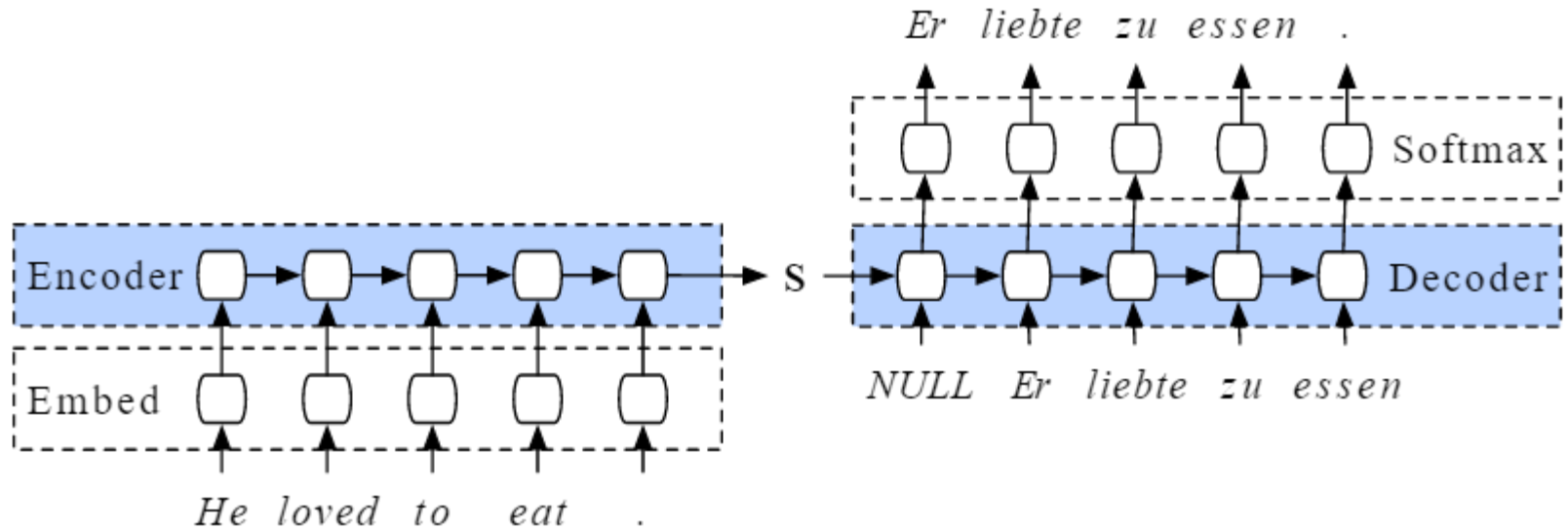
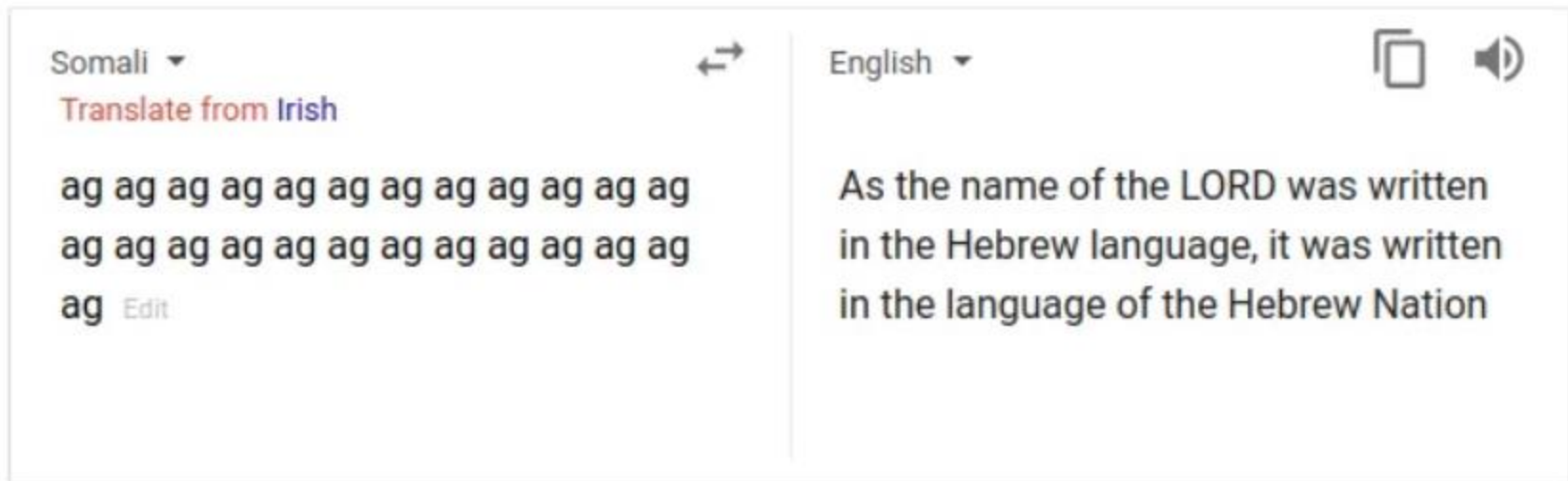


Image taken for smerity.com (latest access: June 2021)

Cons of Neural Machine Translation

- Hard to debug
- Hard to control
- Out-of-vocabulary words
- Domain mismatch from train to test data
- Low-resource language pairs
- Pronoun resolution errors
- Morphological agreement errors

Hard to control



[Open in Google Translate](#)

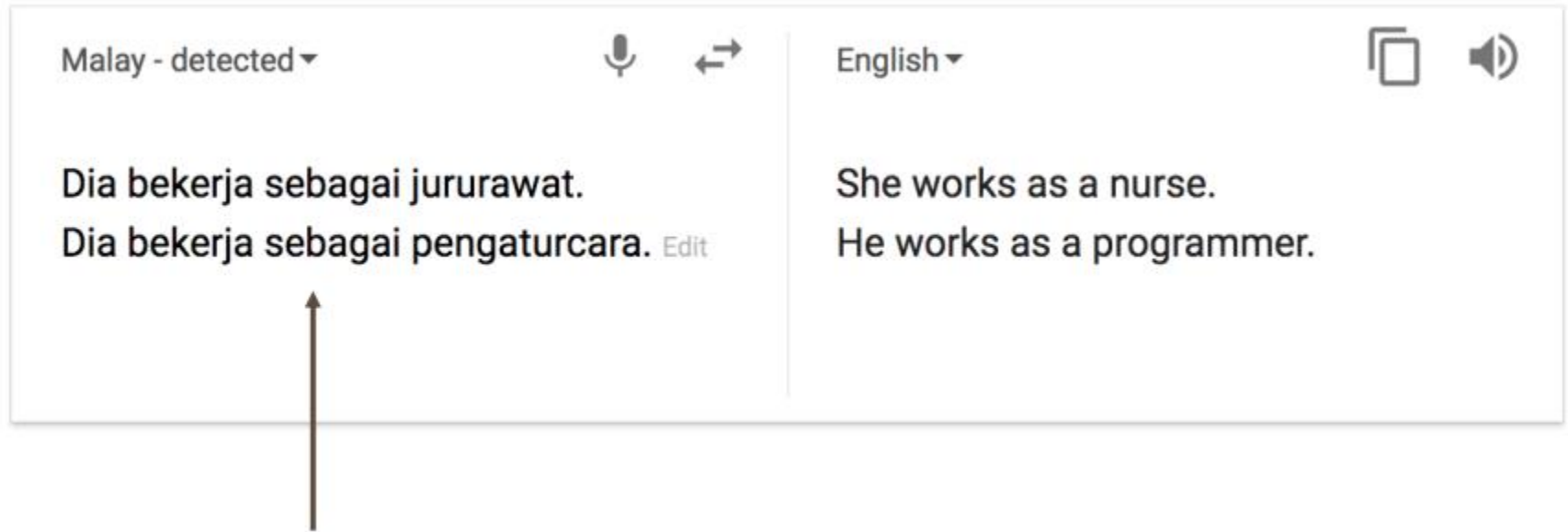
[Feedback](#)

Common sense



?

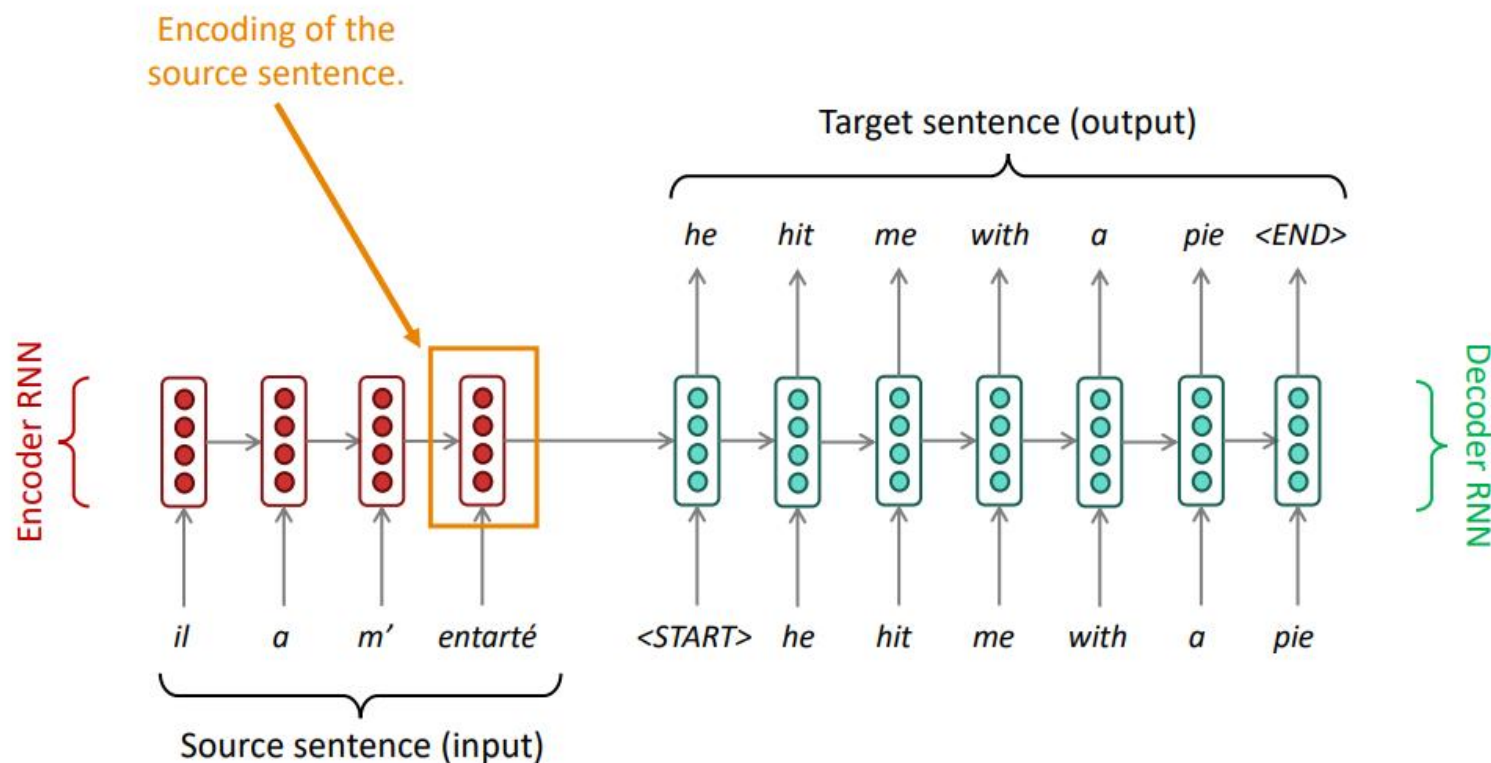
Gender bias



Didn't specify gender

Transformers in Neural Machine Translation

- Encoding of the source allows us to capture all information about the source sentence
 - Given a set of vector values, and a vector query, attention is a technique to compute a weighted sum of the values, dependent on the query



Attention in a nutshell

- Given a set of vector values, and a vector query, attention is a technique to compute a weighted sum of the values, dependent on the query
- The weighted sum is a selective summary of the information contained in the values, where the query determines which values to focus on.
- Attention is a way to obtain a fixed-size representation of an arbitrary set of representations (the values), dependent on some other representation (the query)

Attention in Seq2Seq (1/3)

- Encoder hidden states:

$$h_1, \dots, h_N \in \mathbb{R}^h$$

- On timestep t , we have decoder hidden state $s_t \in \mathbb{R}^h$
- The attention score on timestep t :

$$e^t = [s_t^T h_1, \dots, s_t^T h_N] \in \mathbb{R}^N$$

Attention in Seq2Seq (2/3)

- To get the attention probability distribution on timestep t

$$\alpha^t = \text{softmax}(\mathbf{e}^t) \in \mathbb{R}^N$$

- The attention output on timestep t is a weighted sum of the encoder hidden states:

$$\mathbf{a}_t = \sum_{i=1}^N \alpha_i^t \mathbf{h}_i \in \mathbb{R}^h$$

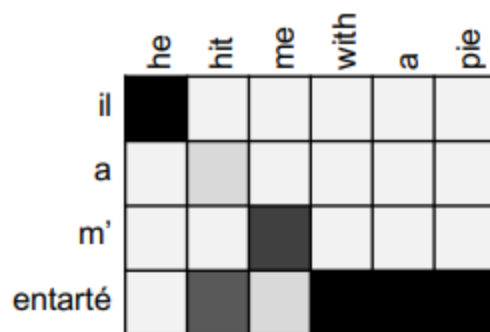
Attention in Seq2Seq (3/3)

- Attention outputs are concatenated with the decoder hidden states as in a traditional seq2seq model

$$[\mathbf{a}_t; \mathbf{s}_t] \in \mathbb{R}^{2h}$$

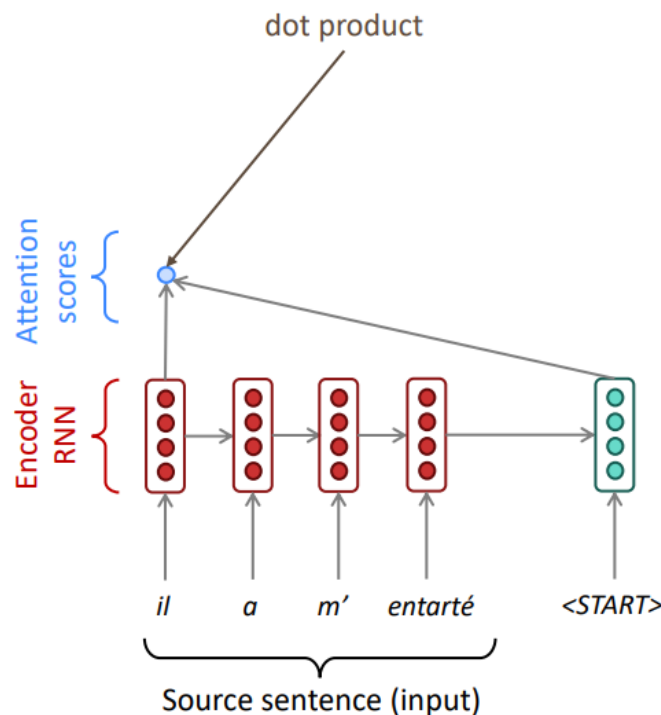
Pros of using attention

- Attention is useful for improving translation performance
 - Focus on certain parts of the source
 - Solve the bottleneck problem
 - the decoder looks directly at source
 - Overcome the vanishing gradient
 - Shortcut to faraway states
 - Saliency map



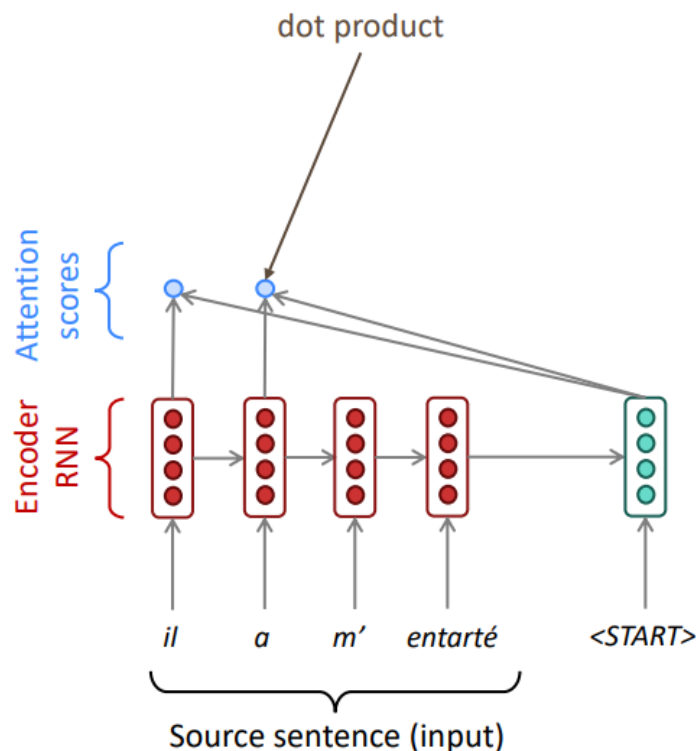
Transformers in Neural Machine Translation

- On each decoder step, attention uses the direct connection to the encoder to attend particular tokens of the source sequence



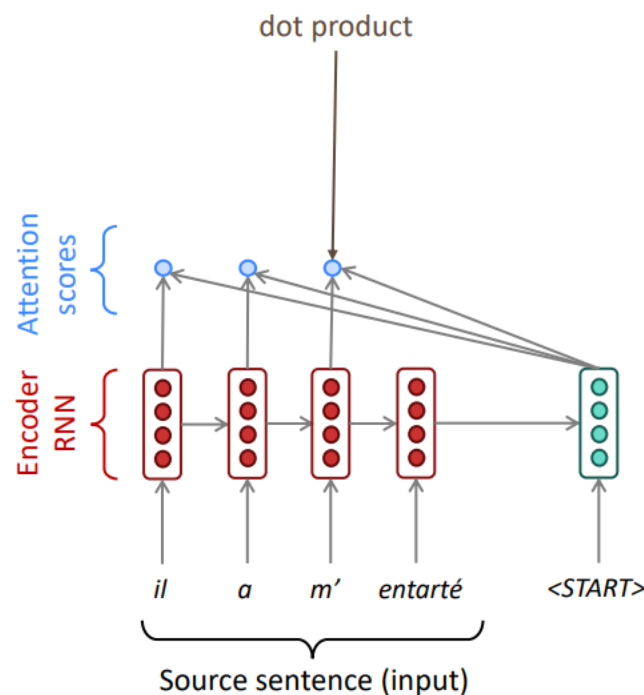
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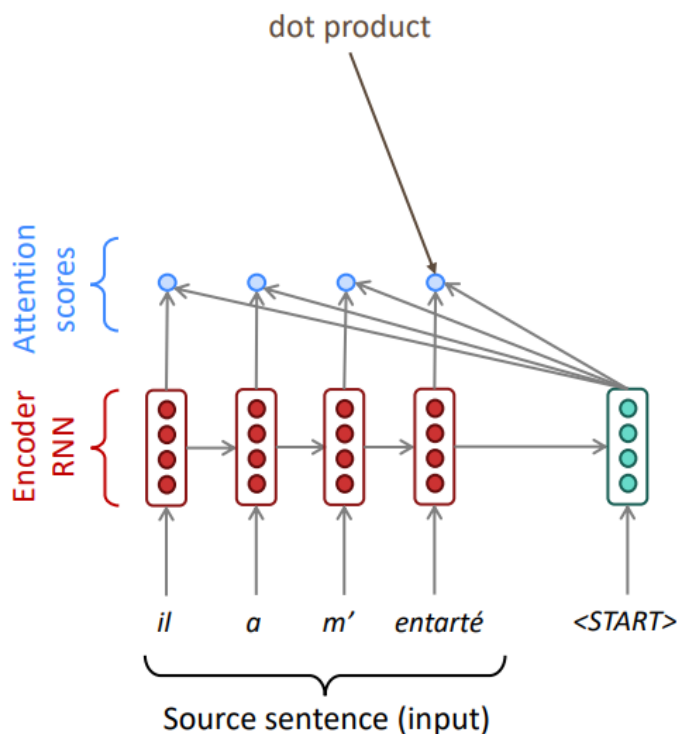
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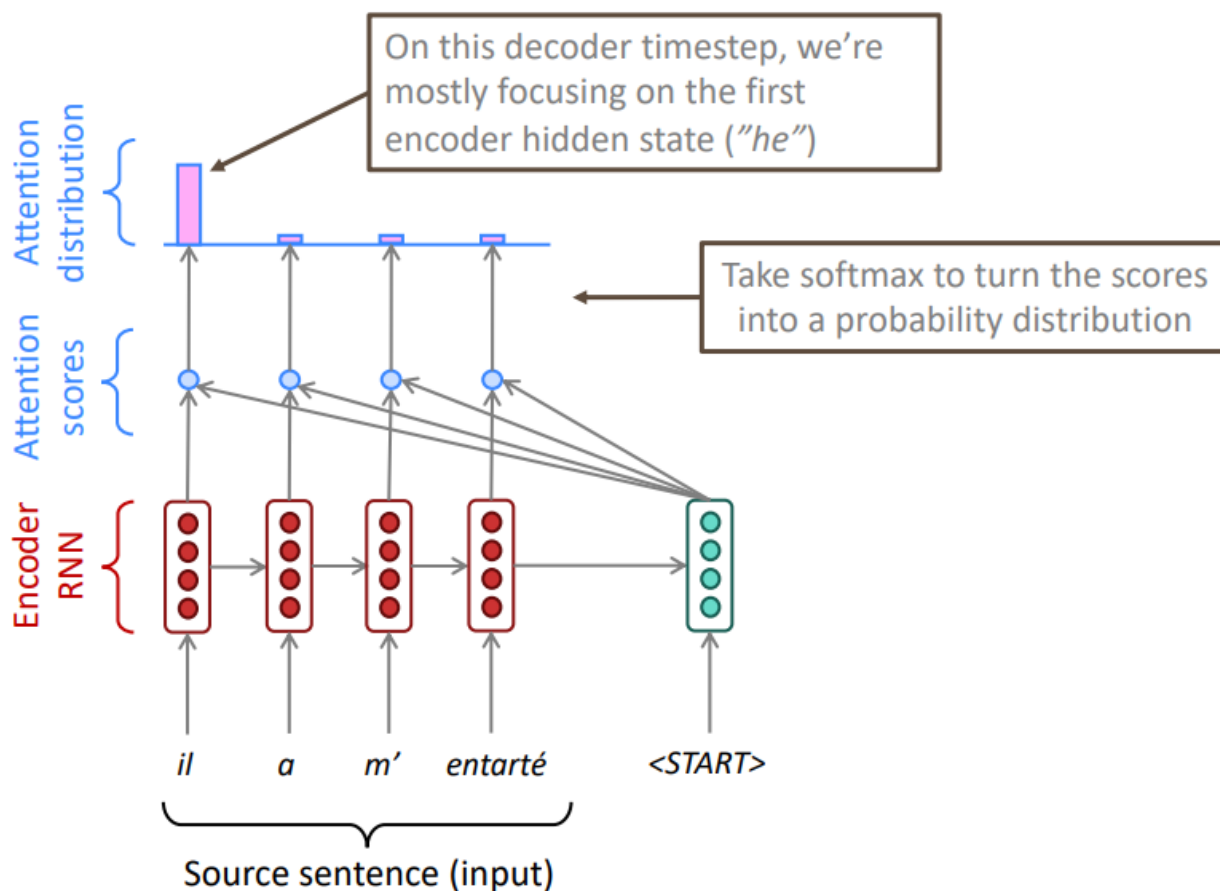


Transformers in Neural Machine Translation

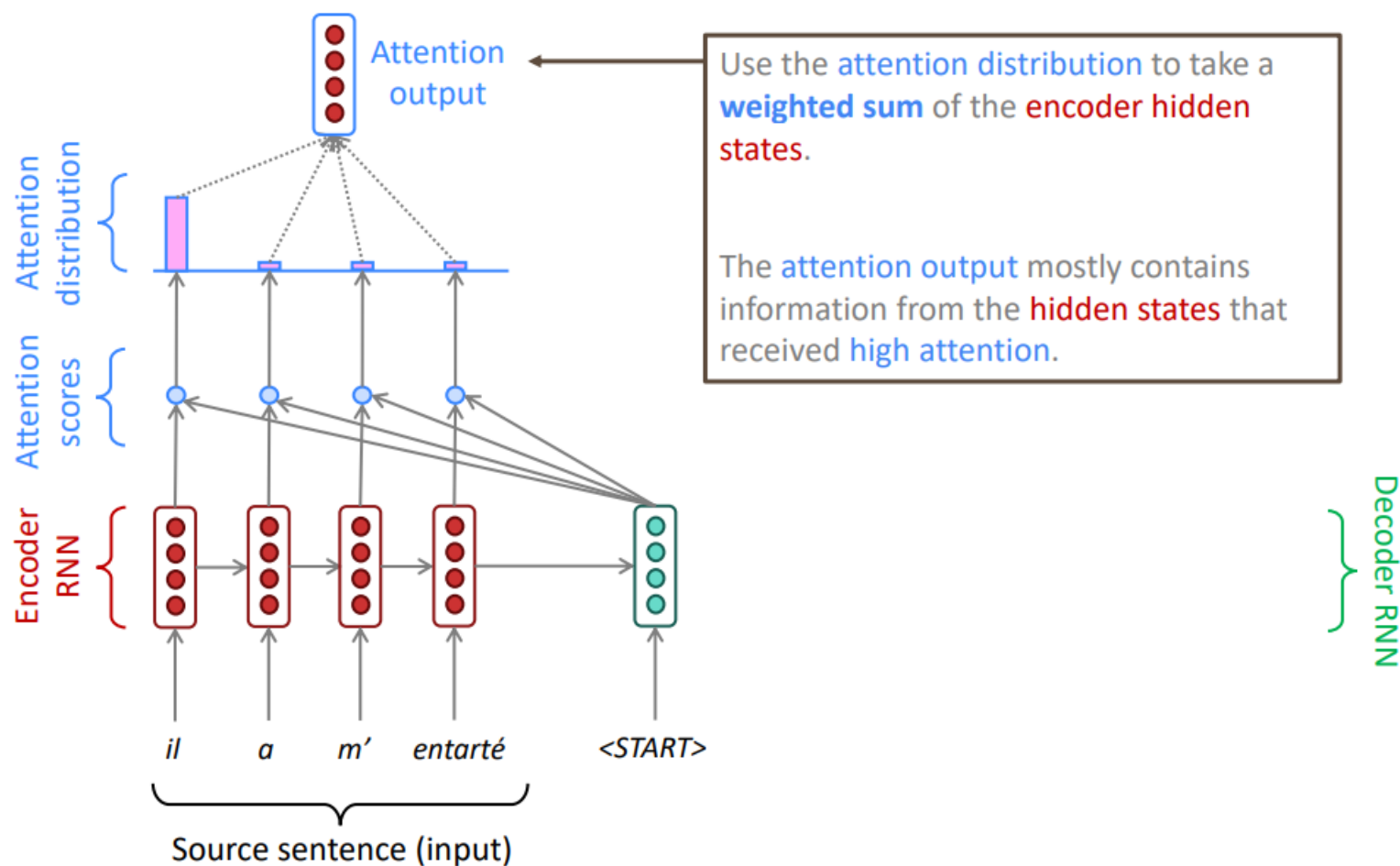
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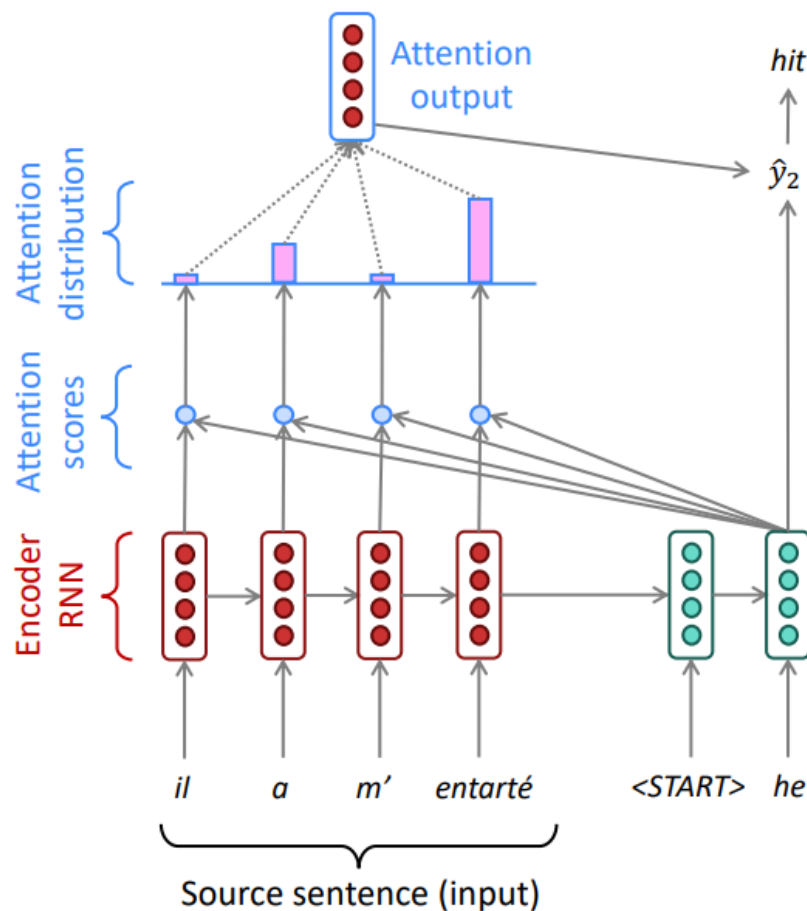
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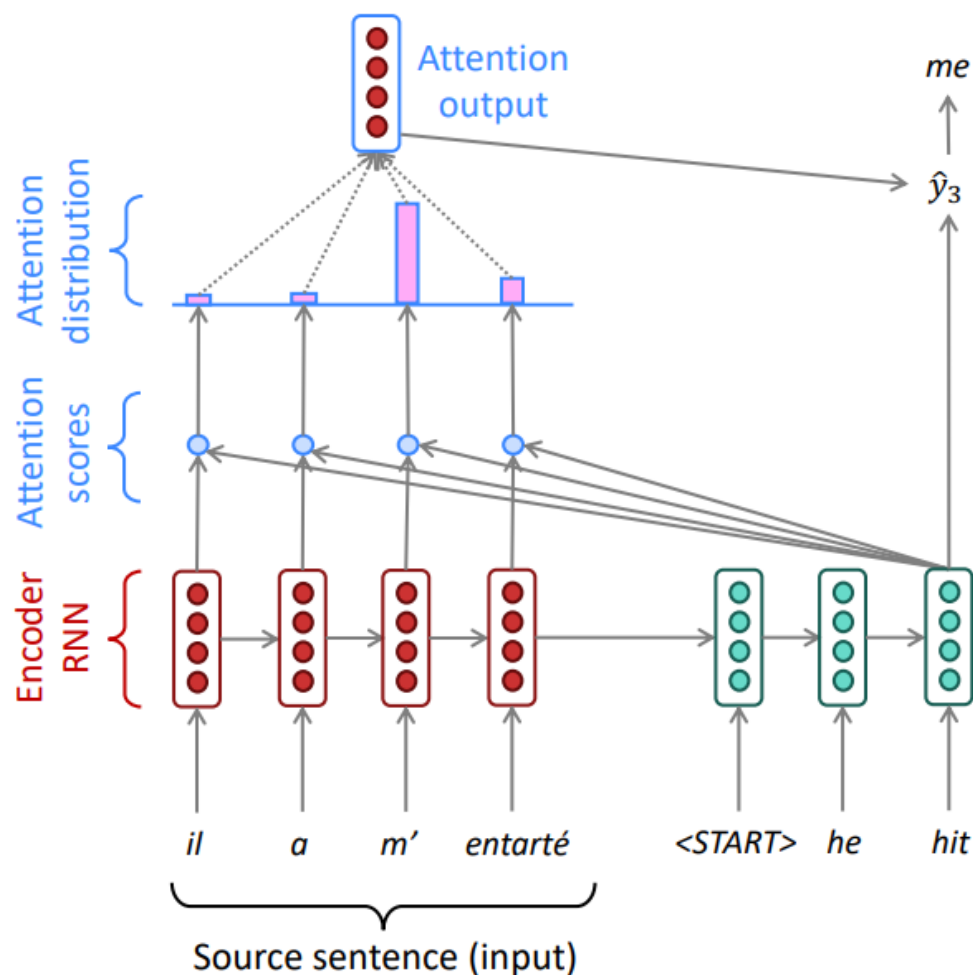
Transformers in Neural Machine Translation



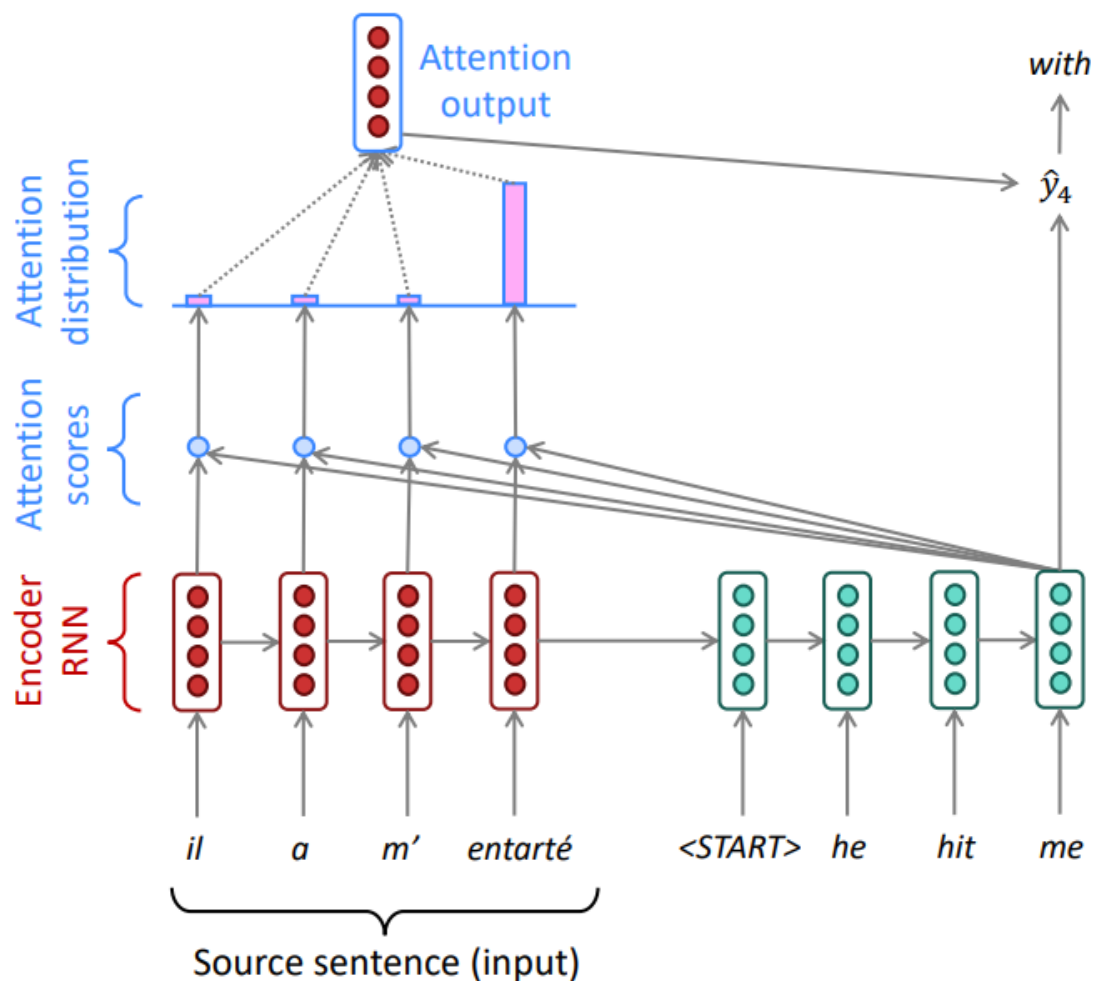
Transformers in Neural Machine Translation



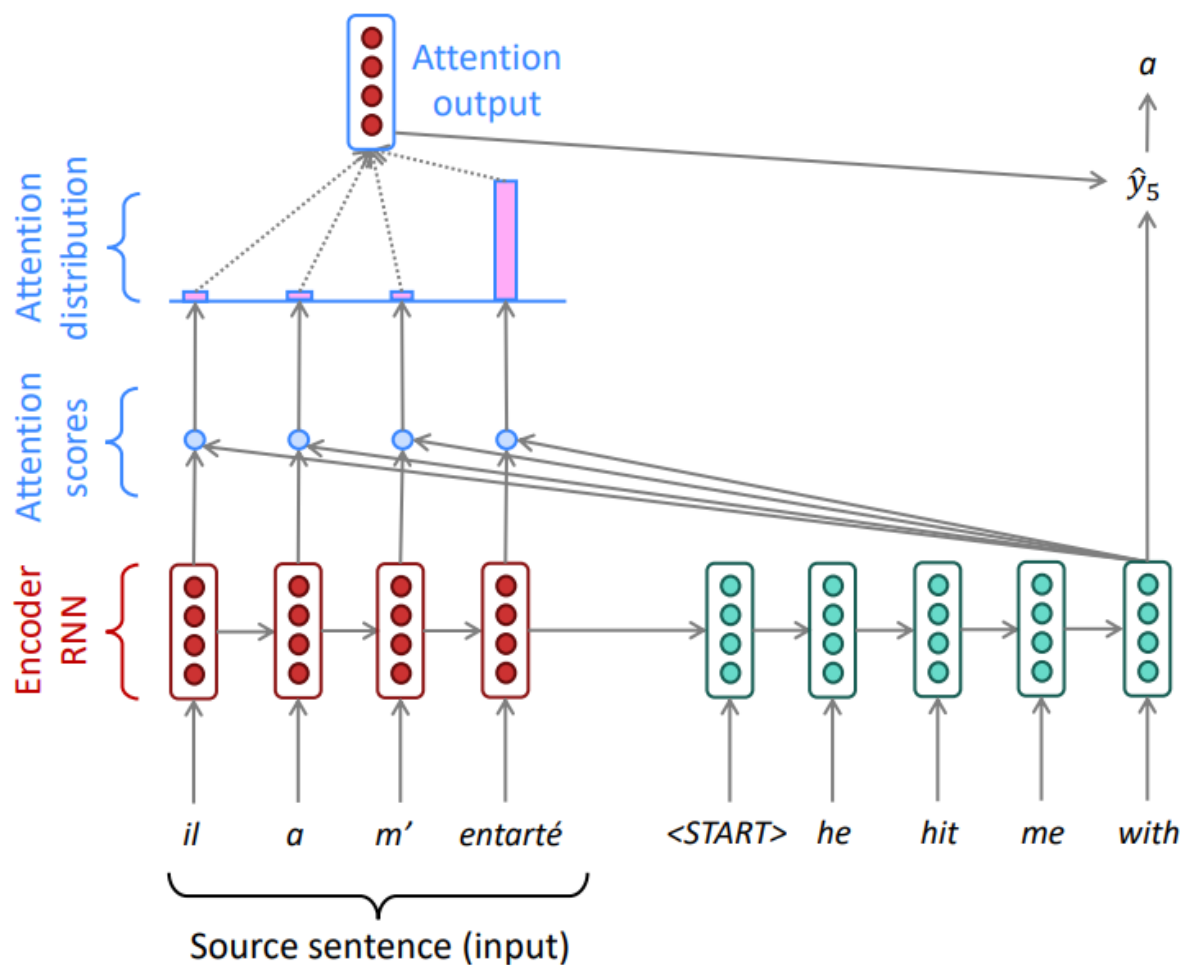
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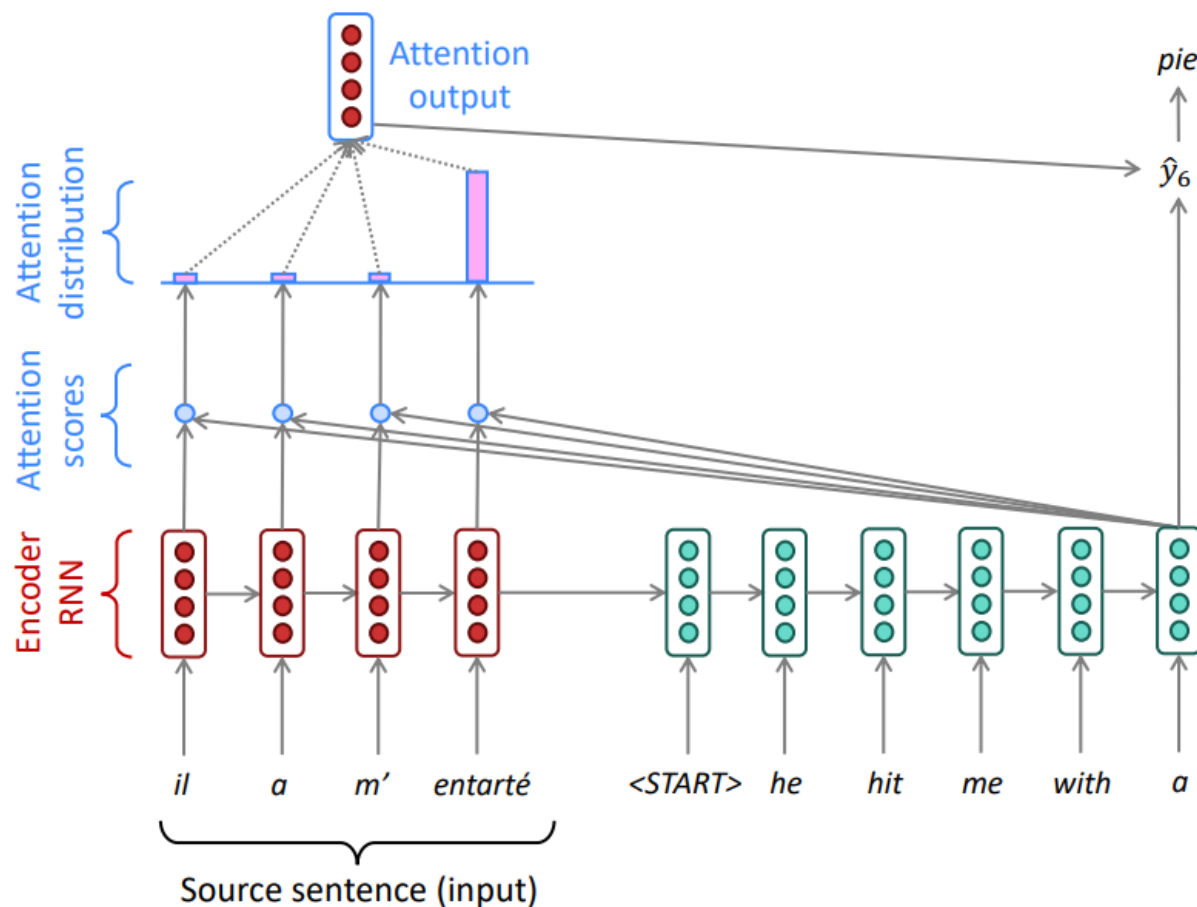
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Transformers in Neural Machine Translation



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BART

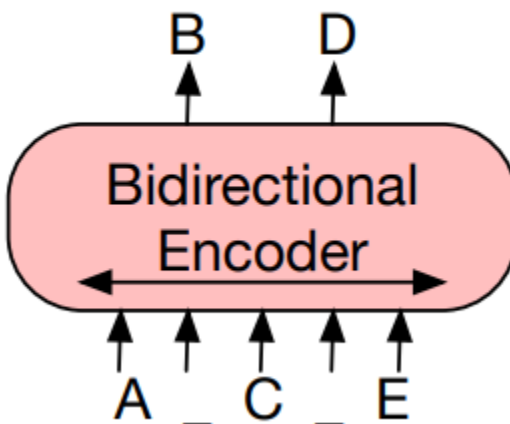
- A state-of-the-art denoising autoencoder for pretraining sequence-to-sequence models
- Based on Transformers
 - Can be seen as a generalization of BERT and GPT
- Trained by
 - corrupting text with arbitrary noising function
 - learning a model to reconstruct the original text

(<https://marian-nmt.github.io/>) (latest access: April 2021)

BERT encoder

- BERT encoder

- Random tokens are replaced with masks, and the document is encoded bidirectionally.
- Missing tokens are predicted independently
 - Not suitable for text generation!

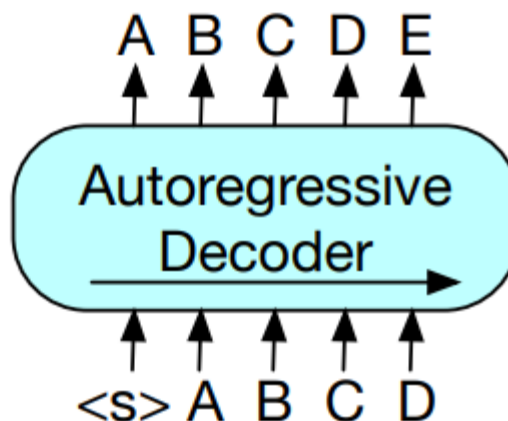


Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, Luke Zettlemoyer.
BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension.

BART

- GPT decoder

- Tokens are predicted auto-regressively
 - Suitable for text generation.
- Words are conditioned on leftward context
 - it cannot learn bidirectional interactions

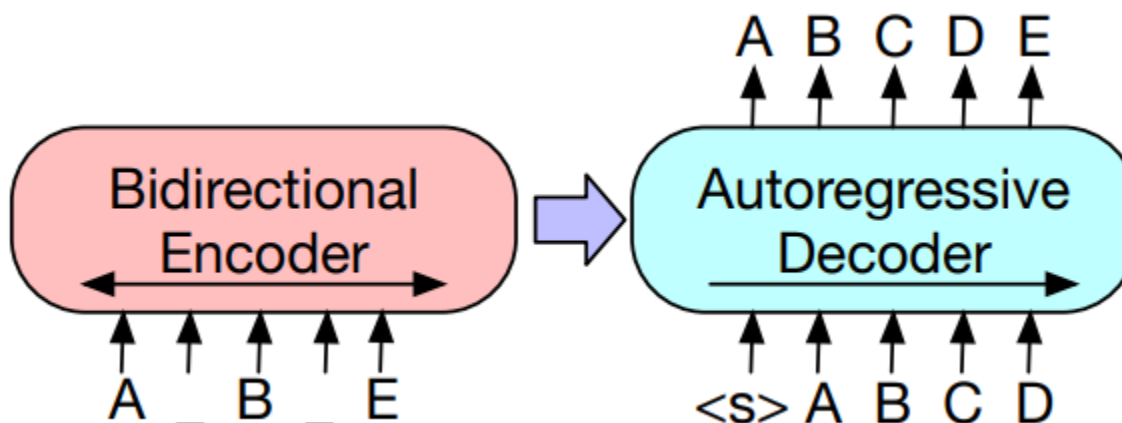


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BART

- BART encoder decoder

- Inputs to the encoder need not be aligned with decoder outputs
 - Arbitrary noise transformations are allowed
- A document is corrupted by replacing spans of text with mask symbols
 - The corrupted document (left) is encoded with a bidirectional model
 - the likelihood of the original document (right) is calculated with an autoregressive decoder

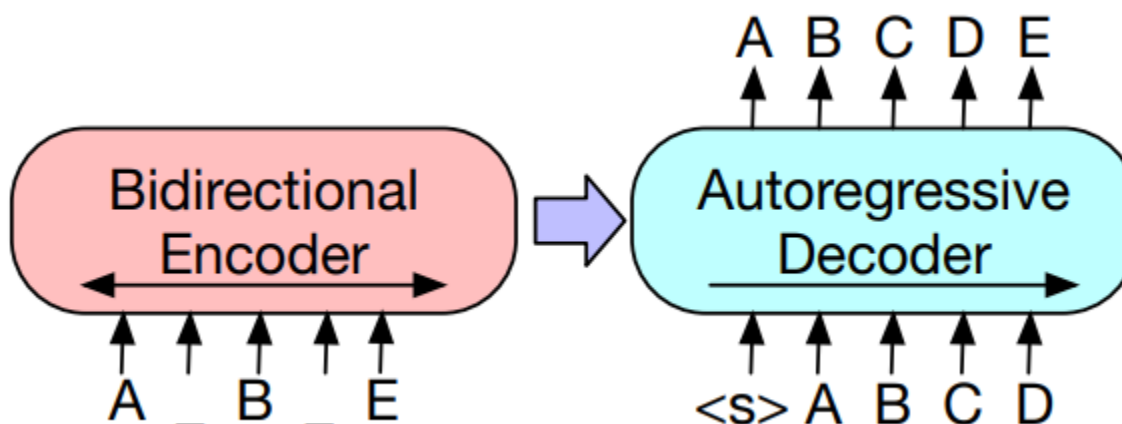


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BART

- BART encoder decoder

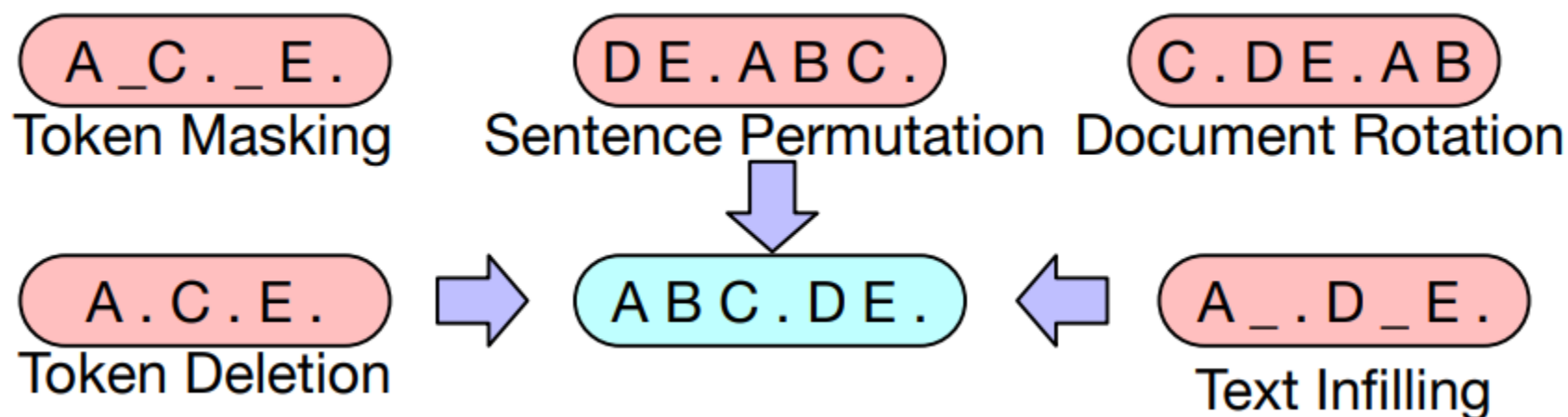
- For fine-tuning, an uncorrupted document is provided as input to both the encoder and decoder, and we use representations from the final hidden state of the decoder



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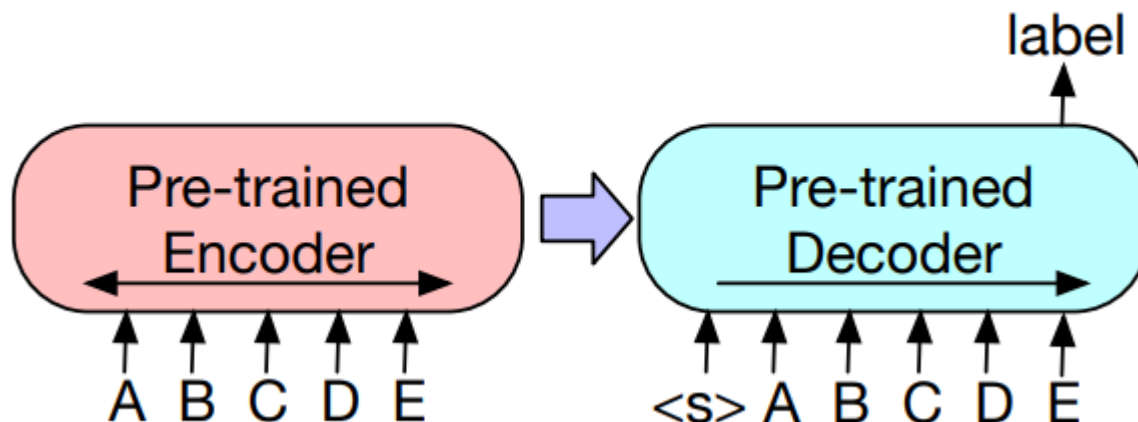
- Transformation for input noising



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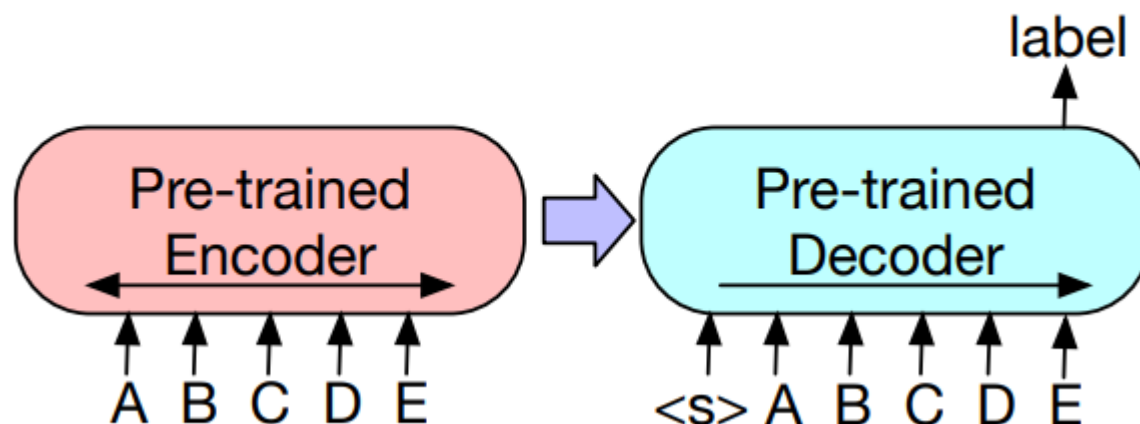
- BART for sequence classification
 - The same input is fed into encoder and decoder
 - The representation from the final output is used



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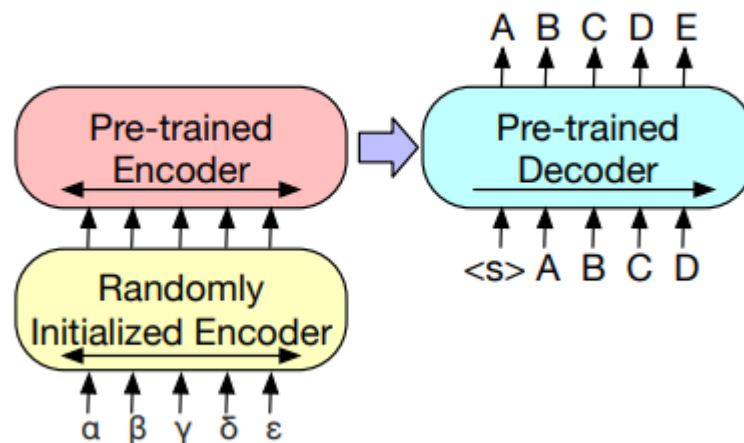
- The same architecture can be exploited for text summarization



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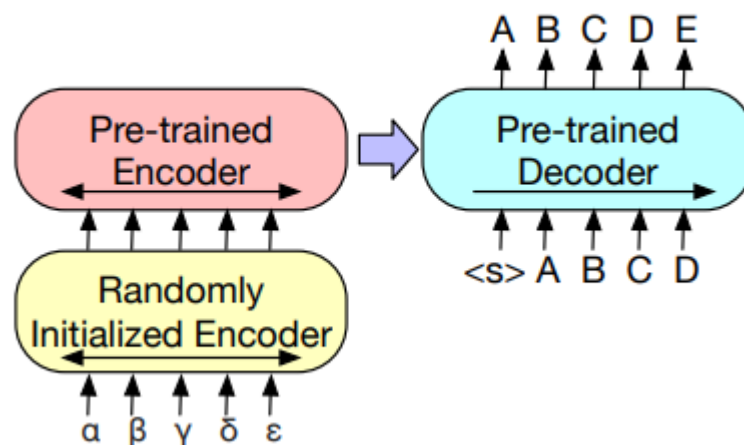
BART

- BART for Machine Translation
 - learn a small additional encoder that replaces the word embeddings in BART
 - The new encoder can use a disjoint vocabulary



BART

- BART for Machine Translation
 - Use the entire BART model (both encoder and decoder) as a single pretrained decoder for machine translation by adding a new set of encoder parameters that are learned from bilingual corpus

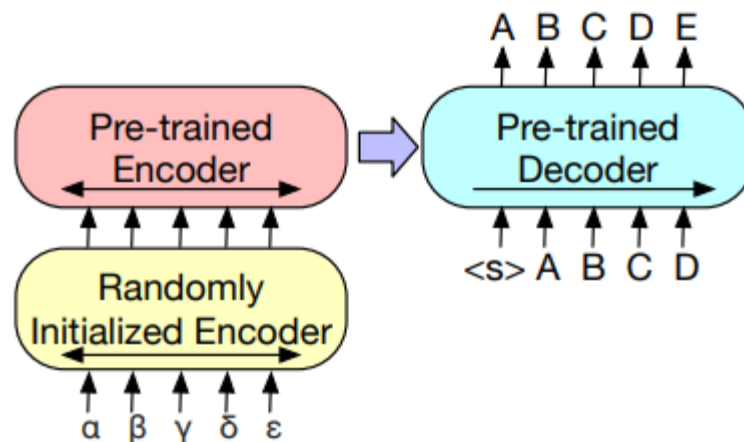


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BART

- BART for Machine Translation

- Replace BERT's encoder embedding layer with a new randomly initialized encoder
- The model is trained end-to-end
 - Train a new encoder to map words in the source language into an input that BART can denoise to the target language.
 - The new encoder can use a separate vocabulary from the original BART model



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Additional reading on BART



- Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, Luke Zettlemoyer. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. CoRR abs/1910.13461 (2019)
- Download and read the paper: <https://arxiv.org/pdf/1910.13461.pdf>

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- Affiliation

- The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
 - <https://dbdmg.polito.it>
 - <https://smartdata.polito.it>

Thank you!